

THE MAGNITUDE CONJECTURE FOR MODULE CATEGORIES

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ABSTRACT. For a finite-dimensional representation-finite algebra over an algebraically closed field, the matrix of Hom dimensions between representatives of the indecomposable modules is known to be invertible. The magnitude of the module category is the sum of the entries of its inverse. We prove that this magnitude is at least the number of isomorphism classes of simple modules, with equality precisely for special biserial algebras. This establishes the conjecture of Børve, Horiatakis, and Kalck. For representation-finite representation-directed algebras, magnitude minus the number of simple modules cannot increase upon taking the quotient by the ideal generated by a primitive idempotent. The proof combines Ringel–Vossieck’s hammock theory, through its description of certain quotient categories as categories of poset representations, with a comparison of almost split sequences. We give direct proofs of the required hammock-theoretic results in an appendix. For general algebras, we reduce to the directed case using finite-dimensional algebras constructed from graded modules.

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Date: September 20, 2026.

2020 Mathematics Subject Classification. 16G10, 16G70.

Key words and phrases. magnitude, representation-finite algebra, Auslander–Reiten quiver, τ -category, special biserial algebra, poset representations.

1. INTRODUCTION

Let A be a finite-dimensional representation-finite algebra over an algebraically closed field k . Let $\text{ind } A$ be a set of representatives of its indecomposable right modules, and fix an ordering of this set. The Hom matrix

$$H_A = (\dim_k \text{Hom}_A(X, Y))_{X, Y \in \text{ind } A}$$

is invertible [4, Lemma 1.9]. Børve, Horiatakis, and Kalck define the *magnitude* of the category $\text{mod } A$ of finite-dimensional right A -modules by

$$\chi(\text{mod } A) = \mathbf{1}^\top H_A^{-1} \mathbf{1},$$

where $\mathbf{1}$ is the column vector whose entries are all one [4, Definitions 1.2 and 1.6]. Thus magnitude is the sum of the entries of the inverse Hom matrix. Reordering $\text{ind } A$ simultaneously permutes the rows and columns of H_A and of H_A^{-1} , leaving this sum unchanged. Magnitude is also invariant under k -linear Morita equivalence (Lemma 2.3).

Write $|A|$ for the number of isomorphism classes of simple modules and $a(A)$ for the number of arrows in the Auslander–Reiten quiver, with $\dim_k \text{Irr}_A(X, Y)$ arrows from X to Y . The formula

$$\chi(\text{mod } A) = 2|\text{ind } A| - |A| - a(A) \quad (1.1)$$

was proved in [4, Lemma 1.9] using almost split sequences. Børve, Horiatakis, and Kalck also proved that $\chi(\text{mod } A) = |A|$ for special biserial algebras (see Definition 4.12) [4, Theorem 3.5]. They conjectured that $\chi(\text{mod } A) \geq |A|$ for every representation-finite algebra, with equality precisely for special biserial algebras [4, Conjecture A]. We prove this conjecture.

Theorem A (= Theorem 5.19). *Let A be a finite-dimensional representation-finite algebra over an algebraically closed field. Write $|A|$ for its number of isomorphism classes of simple modules. Then the following hold.*

- (1) $\chi(\text{mod } A) \geq |A|$.
- (2) $\chi(\text{mod } A) = |A|$ if and only if A is special biserial.

The proof begins with representation-directed algebras. We show that the *magnitude surplus* $\sigma(A) = \chi(\text{mod } A) - |A|$ satisfies

$$\sigma(A) \geq \sigma(A/AeA)$$

for every primitive idempotent e . The quotient remains representation-finite and representation-directed, since its module category is a full subcategory of $\text{mod } A$. Successive deletions end at the zero algebra, whose surplus is zero, and hence prove nonnegativity. A central step is an equivalence between the additive quotient

$$(\text{mod } A)/[\text{mod}(A/AeA)]$$

and a category of *representations of a finite poset by subspaces*. Here A/AeA -modules are regarded as A -modules annihilated by e , and brackets denote the ideal of maps factoring through them. A representation of a poset by subspaces consists of a finite-dimensional space V with subspaces V_t , one for each poset element, such that $V_s \subseteq V_t$ when $s \leq t$; morphisms preserve the specified subspaces (Definition 3.15). This equivalence is due to Ringel–Vossieck [11, Section 10, Theorem, p. 243]. Their hammock theory relates the composition multiplicities of a fixed simple module to poset representations and the lengths of paths in their Auslander–Reiten quivers. We describe the poset and the equivalence explicitly in Construction 3.16 and Proposition 3.17. Their path-length inequality [11, Section 8, Proposition, p. 237] says that every path from the unique source to the unique sink has length at least the size of the poset. It gives the bound on the number of irreducible arrows needed here. Iyama’s quotient theorem (Proposition 3.5) obtains the quotient’s τ -sequences from almost split sequences by deleting the summands annihilated by e . These sequences generalize almost split sequences to additive categories. We combine the arrow bound with a comparison of almost split sequences over A and A/AeA (Proposition 4.10) to prove the deletion inequality $\sigma(A) \geq \sigma(A/AeA)$. Equality in this deletion inequality forces the simple module corresponding to e to occur at most once as a composition factor of each indecomposable.

Appendix A proves the structural statements used here in their stated form, using the module and τ -sequence preliminaries of this paper. Thus these parts of the argument can be read without

consulting the hammock literature. The proofs establish the multiplicity bounds, realize prescribed poset representations, and prove the path-length inequality and its equality case.

For a general algebra A , we choose a *standard form* S (Theorem 5.1). Its Auslander–Reiten quiver is isomorphic to that of A , preserving projective vertices, so $\sigma(S) = \sigma(A)$. The standard form admits an integer grading for which every indecomposable S -module admits a graded module structure (Construction 5.2 and Proposition 5.6). For each $m \geq 0$, we construct a representation-finite representation-directed algebra S_m whose modules correspond to graded S -modules supported in degrees $0, \dots, m$ (Construction 5.7). Counting its indecomposables and arrows gives $\sigma(S_m) = (m+1)\sigma(A) + O(1)$, with error bounded independently of m . Dividing by $m+1$ and letting $m \rightarrow \infty$, we obtain $\sigma(A) \geq 0$ from $\sigma(S_m) \geq 0$. For the equality case, we realize arbitrarily many separated copies of a fixed S_m as a quotient of a larger interval algebra. The deletion inequality and the same counting formula then imply that $\sigma(A) = 0$ forces every S_m to have surplus zero (Proposition 5.12). Almost split sequences over a sufficiently long interval algebra then imply that A is special biserial. Figure 1 gives the principal implications and their references.

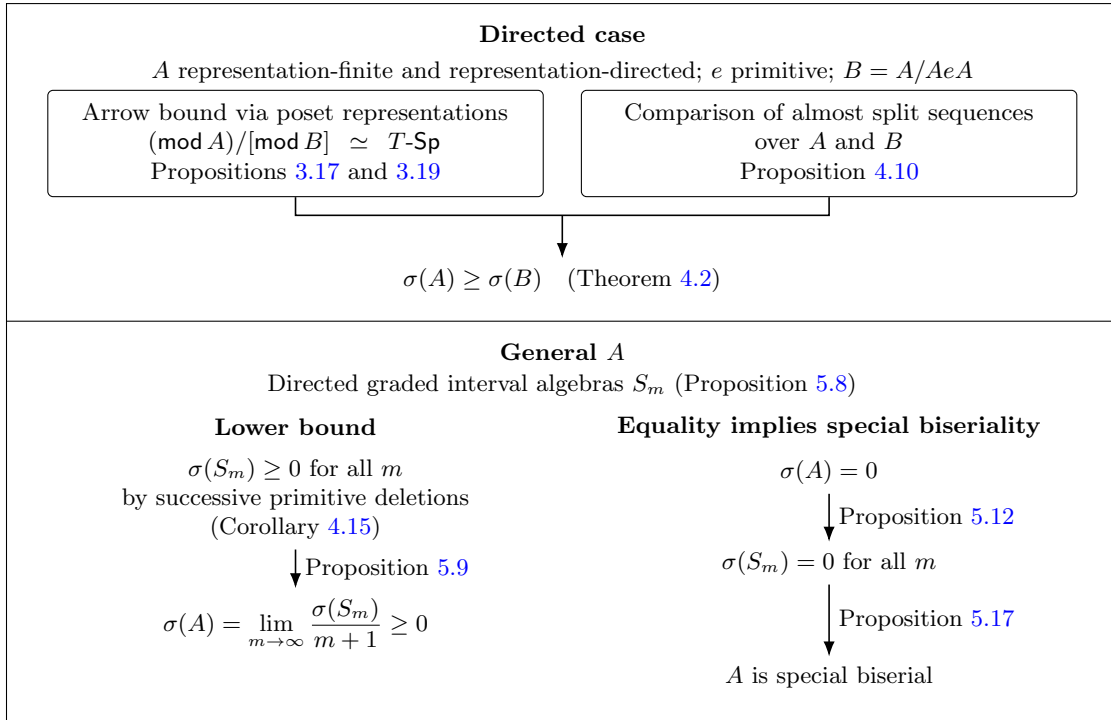


FIGURE 1. Primitive deletion, the lower bound, and the equality case. The poset equivalence and arrow bound follow from Ringel–Vossieck [11]. Here $\sigma(A) = \chi(\text{mod } A) - |A|$, $T\text{-Sp}$ denotes representations of a finite poset T by subspaces, S is a standard form of A , and $\text{mod } S_m$ is equivalent to the category of graded S -modules supported in degrees $0, \dots, m$ (Proposition 5.8).

Organization. Section 2 recalls the magnitude formula, almost split sequences, and their generalization to τ -categories. For a directed algebra, Section 3 describes the additive quotient and its τ -sequences using Iyama’s quotient theorem (Proposition 3.5). It gives the poset representation equivalence and deduces an arrow bound from Ringel–Vossieck’s results. Section 4 combines this bound with a comparison of almost split sequences to prove Theorem 4.2. Section 5 constructs finite algebras from graded modules over the standard form and applies the directed result to prove both assertions of Theorem A. Appendix A proves the structural results on the quotient directly and gives the detailed example for the Dynkin quiver of type D_4 (Example A.2).

Conventions and notation. Throughout, k is algebraically closed and algebras are finite-dimensional k -algebras. Modules are finite-dimensional right modules, and $\mathbf{mod} A$ denotes their category. We write $D_k = \mathrm{Hom}_k(-, k)$, and add M for direct summands of finite direct sums of copies of M . For a Krull–Schmidt category $\mathcal{C}$, $\mathrm{ind} \mathcal{C}$ is a chosen set of indecomposable representatives, also regarded as the full subcategory on those objects. We write $\mathcal{C}(X, Y) = \mathrm{Hom}_{\mathcal{C}}(X, Y)$ and abbreviate $\mathrm{ind}(\mathbf{mod} A)$ to $\mathrm{ind} A$. A category is representation-finite if it has finitely many indecomposable isomorphism classes. The categorical radical is $\mathrm{rad}_{\mathcal{C}}$, and $\mathrm{Irr}_{\mathcal{C}}(X, Y) = \mathrm{rad}_{\mathcal{C}}(X, Y) / \mathrm{rad}_{\mathcal{C}}^2(X, Y)$. The *irreducible quiver* of $\mathcal{C}$ has vertices $\mathrm{ind} \mathcal{C}$ and $\dim_k \mathrm{Irr}_{\mathcal{C}}(X, Y)$ arrows from X to Y . We write $\mathrm{indeg}_{\mathcal{C}}(X)$ and $\mathrm{outdeg}_{\mathcal{C}}(X)$ for the numbers of incoming and outgoing arrows, including multiplicities. The *Auslander–Reiten translation quiver* Γ_A is this quiver of $\mathbf{mod} A$, equipped with Auslander–Reiten translation and with its projective and injective vertices marked. Translations are displayed by dotted arrows $X \dashrightarrow \tau_A X$; they do not count as arrows in paths or arrow counts. An algebra is *representation-directed* if there is no cycle of nonzero nonisomorphisms between indecomposable modules. We also use *directed* in this sense for a category. Composition is written gf for first f , then g . We use the same order for paths: $\beta\alpha$ traverses α first, then β .

Use of AI. The research was conducted under the author’s direction using GPT-5.6-Sol, Claude Fable 5, Claude Opus 5, and GPT-6-Astra. GPT-5.6-Sol and Claude Fable 5 developed the initial proof, and GPT-5.6-Sol produced earlier manuscript drafts. GPT-6-Astra subsequently developed the simplified proof presented here and wrote a new manuscript based on it, retaining selected arguments and examples from the earlier drafts. Claude Opus 5 extensively reviewed the proofs and successive manuscript drafts. The author directed the research and revisions and set the structure and much of the wording of the present text. The author is responsible for the mathematical content and the final manuscript.

2. MAGNITUDE AND ALMOST SPLIT SEQUENCES

The magnitude formula expresses the problem in terms of the numbers of indecomposable modules, simple modules, and arrows in the Auslander–Reiten quiver. We recall this formula and the properties of almost split sequences used in the quotient argument.

2.1. The counting formula. Fix a representation-finite algebra A .

Definition 2.1. Put

$$a(A) = \sum_{X, Y \in \mathrm{ind} A} \dim_k \mathrm{Irr}_A(X, Y), \quad \sigma(A) = \chi(\mathbf{mod} A) - |A|.$$

We call $\sigma(A)$ the *magnitude surplus*.

The following formula expresses magnitude in terms of these counts.

Lemma 2.2 ([4, Lemma 1.9]). *The matrix $H_A = (\dim_k \mathrm{Hom}_A(X, Y))_{X, Y \in \mathrm{ind} A}$ is invertible, and*

$$\chi(\mathbf{mod} A) = 2|\mathrm{ind} A| - |A| - a(A).$$

Consequently,

$$\sigma(A) = 2(|\mathrm{ind} A| - |A|) - a(A). \quad (2.1)$$

We use Morita equivalence to reduce to basic algebras and additivity to count the surplus of products of interval algebras in Section 5.

Lemma 2.3. *Magnitude, $|A|$, and $\sigma(A)$ are Morita invariant and additive on products of algebras.*

Proof. A k -linear Morita equivalence preserves simple classes, indecomposable classes, and Hom dimensions. For a product of algebras, the Hom matrix is block diagonal, so its inverse is block diagonal and its entry sum is the sum for the factors. The simple classes are the disjoint union of those of the factors. Subtracting their numbers proves the assertion for σ . $\square$

Remark 2.4. We call passage from A to A/AeA , for a primitive idempotent e , *primitive deletion*. To study it over a basic algebra, choose an idempotent f with $fe = ef = e$, $AfA = A$, and fAf basic. From a complete orthogonal set of primitive idempotents containing e , take one for each isomorphism class of indecomposable projectives, including e , and let f be their sum. The idempotent e remains primitive in fAf , since $e(fAf)e = eAe$ is local. Since $(Xf)e = Xe$, the Morita equivalence $X \mapsto Xf$ restricts to

$$\text{mod}(A/AeA) \simeq \text{mod}(fAf/(fAf)e(fAf)).$$

Indeed, $f(AeA)f = (fAf)e(fAf)$. Writing $\bar{f}$ for the image of f in $B = A/AeA$, the equality $AfA = A$ gives $B\bar{f}B = B$, so $\bar{f}$ is full and induces the displayed Morita equivalence. By Lemma 2.3, this reduction preserves $\sigma(A) - \sigma(A/AeA)$.

2.2. Almost split maps and τ -sequences. We recall almost split maps and sequences, then define the τ -sequences used in the additive quotient.

A map $f : V \rightarrow X$ to an indecomposable X is *right almost split* if it is not a retraction and every non-retraction to X factors through it. It is *right minimal* if every $u \in \text{End}(V)$ with $fu = f$ is invertible. Left almost split maps and left minimality are defined dually, using sections.

Definition 2.5. For a nonprojective indecomposable X , write its almost split sequence as

$$0 \longrightarrow \tau_A X \longrightarrow \theta_A X \longrightarrow X \longrightarrow 0.$$

At a projective X , put $\tau_A X = 0$, $\theta_A X = \text{rad } X$, and use the radical inclusion $\theta_A X \rightarrow X$. Dually, write the sequence starting at a noninjective X as

$$0 \longrightarrow X \longrightarrow \theta_A^- X \longrightarrow \tau_A^- X \longrightarrow 0,$$

and put $\theta_A^- X = X/\text{soc } X$, $\tau_A^- X = 0$ at an injective X . Extend $\tau_A, \tau_A^-, \theta_A, \theta_A^-$ additively to arbitrary modules, as assignments on isomorphism classes.

The following standard facts relate almost split maps to the arrows of Γ_A . For an indecomposable Y , write $[V : Y]_\oplus$ for its multiplicity as a direct summand of V .

Proposition 2.6 ([2, V, Section 1; VII, Proposition 1.3]). *For indecomposable A -modules X, Y , the maps $\theta_A X \rightarrow X$ are minimal right almost split and the maps $X \rightarrow \theta_A^- X$ are minimal left almost split. Their middle multiplicities satisfy*

$$[\theta_A X : Y]_\oplus = \dim_k \text{Irr}_A(Y, X), \quad [\theta_A^- X : Y]_\oplus = \dim_k \text{Irr}_A(X, Y). \quad (2.2)$$

The map τ_A is a bijection from nonprojective to noninjective indecomposables, with inverse τ_A^- .

We will use the following form of Auslander–Reiten duality.

Proposition 2.7 ([1, IV, Theorem 2.13]). *For modules U, V , there are natural isomorphisms*

$$D_k \text{Ext}_A^1(U, V) \cong \underline{\text{Hom}}_A(\tau_A^- V, U) \cong \overline{\text{Hom}}_A(V, \tau_A U). \quad (2.3)$$

Here $\underline{\text{Hom}}$ is *Hom modulo maps through projectives*, and $\overline{\text{Hom}}$ is *Hom modulo maps through injectives*. The formulas include the zero cases when V is injective or U is projective.

An additive quotient need not be an abelian category. To describe the images of almost split sequences in such a quotient, we use weak kernels and weak cokernels, which we now recall.

Definition 2.8. In an additive category, a map $f : L \rightarrow V$ is a *weak kernel* of $g : V \rightarrow Y$ if $gf = 0$ and every map $h : Z \rightarrow V$ with $gh = 0$ factors through f . It is minimal if $fu = f$ forces $u \in \text{End}(L)$ to be invertible. Weak cokernels and their minimality are defined dually.

Definition 2.9 ([7, Section 1.3]). Let $\mathcal{C}$ be a Krull–Schmidt category. A complex $L \xrightarrow{f} V \xrightarrow{g} Y$ is a *right τ -sequence* if f, g are radical, f is a minimal weak kernel of g , and for every object Z ,

$$\text{Im}(\mathcal{C}(V, Z) \xrightarrow{- \circ f} \mathcal{C}(L, Z)) = \text{rad}_{\mathcal{C}}(L, Z), \quad \text{Im}(\mathcal{C}(Z, V) \xrightarrow{g \circ -} \mathcal{C}(Z, Y)) = \text{rad}_{\mathcal{C}}(Z, Y).$$

A *left τ -sequence* satisfies the same radical and image conditions, with g a minimal weak cokernel of f . The category $\mathcal{C}$ is a τ -category if every indecomposable is the last term of a right τ -sequence and the first term of a left τ -sequence.

Since f, g are radical, their composites with arbitrary maps are radical. The two image equalities therefore say that every radical map out of L factors through f , and every radical map into Y factors through g .

Remark 2.10 ([8, Sections 2.1 and 2.3]). At an indecomposable last term, the last map of a right τ -sequence is minimal right almost split, and the first term is either zero or indecomposable. Dually, the first map of a left τ -sequence starting at an indecomposable is minimal left almost split, and its last term is either zero or indecomposable. A right τ -sequence is determined up to isomorphism by its indecomposable last term, and a left τ -sequence by its indecomposable first term. A right τ -sequence with nonzero indecomposable endpoints is also a left τ -sequence, and conversely. Thus taking the first term of a right τ -sequence gives a bijection from the indecomposables whose right τ -sequence has nonzero first term to those whose left τ -sequence has nonzero last term; its inverse takes the last term of the left τ -sequence.

Example 2.11. The category $\mathbf{mod} A$ is a τ -category: its right and left τ -sequences are the almost split sequences with their outer zeros omitted, together with $0 \rightarrow \text{rad } P \rightarrow P$ at projectives and $I \rightarrow I/\text{soc } I \rightarrow 0$ at injectives. The almost split factorization properties give the two image equalities in Definition 2.9; the ordinary kernels and cokernels give the required minimal weak kernels and cokernels.

We will repeatedly express radical maps as sums of composites of irreducible maps.

Lemma 2.12. *Let $\mathcal{C}$ be a Hom-finite Krull–Schmidt k -category with finitely many indecomposable isomorphism classes. Choose maps representing a basis of each $\text{Irr}_{\mathcal{C}}(X, Y)$. Then its radical is nilpotent, and every radical map between indecomposables is a linear combination of nonempty paths in the chosen maps.*

Proof. Put $M = \bigoplus_{X \in \text{ind } \mathcal{C}} X$ and $\Lambda = \text{End}_{\mathcal{C}}(M)$. The spaces $\text{rad}_{\mathcal{C}}(X, Y)$ are the corresponding matrix blocks of $\text{rad } \Lambda$. Since Λ is finite-dimensional, choose N with $(\text{rad } \Lambda)^N = 0$; then $\text{rad}_{\mathcal{C}}^N = 0$. Modulo $\text{rad}_{\mathcal{C}}^2$, every radical map is a linear combination of the chosen representatives. Repeating this expansion in each factor expresses it modulo $\text{rad}_{\mathcal{C}}^{n+1}$ as a sum of paths of lengths at most n . Taking $n = N - 1$ proves the claim. $\square$

3. THE QUOTIENT AND ITS POSET REPRESENTATIONS

For a directed algebra and a primitive idempotent e , we form an additive quotient whose indecomposable objects are the indecomposable A -modules X with $Xe \neq 0$. Iyama’s quotient theorem describes its τ -sequences, and Ringel–Vossieck’s results identify it with a category of poset representations. We state these results in terms of the quotient and use them to bound the number of its irreducible arrows. This bound will give one of the nonnegative contributions to the change in magnitude under deletion. The proofs of the multiplicity, height, and poset results are given in Appendix A.

Setting 3.1. Throughout Sections 3–4 and Appendix A, let A be basic, representation-finite, and representation-directed. Fix a primitive idempotent e , and put

$$J = AeA, \quad B = A/J, \quad \mathcal{D} = (\mathbf{mod} A)/[\mathbf{mod} B], \quad P = eA, \quad E = P/\text{rad } P, \quad I = D_k(Ae).$$

Thus P is the projective cover of the simple module E , and I is its injective envelope. The quotient has the same objects as $\mathbf{mod} A$; its morphisms are taken modulo maps factoring through B -modules. We regard $\mathbf{mod} B$ as the full subcategory of modules X with $Xe = 0$, by restriction of scalars along $A \twoheadrightarrow B$. Put $d_X = \dim_k Xe = [X : E]$, where $[X : E]$ is composition multiplicity. The equality follows by applying the exact functor $(-)e$ to a composition series.

Lemma 3.2. *The subcategory $\mathbf{mod} B$ of $\mathbf{mod} A$ is closed under submodules, quotients, extensions and direct summands. The algebra B is representation-finite and representation-directed. For every $W \in \mathbf{mod} B$, $\text{Hom}_A(E, W) = \text{Hom}_A(W, E) = 0$.*

Proof. Submodules and quotients of modules annihilated by J are again annihilated by J . For extension closure, let $0 \rightarrow X \rightarrow V \rightarrow Y \rightarrow 0$ be exact with $XJ = YJ = 0$. Then $VJ \subseteq X$. Since $e^2 = e$, the ideal $J = AeA$ satisfies $J^2 = J$, so $VJ = VJ^2 \subseteq XJ = 0$. Moreover, $\text{mod } B$ is a full subcategory of $\text{mod } A$ closed under direct summands. The inclusion preserves indecomposables and their Hom spaces, so B is representation-finite and directed. Finally, $We = 0$ implies $[W : E] = 0$, so E is neither a submodule nor a quotient of W . This proves both Hom vanishings. $\square$

3.1. Evaluation and irreducible maps in the quotient. We first show that the indecomposable objects of $\mathcal{D}$ are precisely the indecomposable A -modules X with $Xe \neq 0$. We also show that a map becomes zero in $\mathcal{D}$ if and only if its restriction at e is zero. For a module map $f : X \rightarrow Y$, write $f_e : Xe \rightarrow Ye$ for its restriction.

Lemma 3.3. *Evaluation at e and passage to $\mathcal{D}$ have the following properties.*

- (1) *The functor $F : \mathcal{D} \rightarrow \text{mod } k$, sending X to Xe and the class of f to f_e , is faithful.*
- (2) *Every indecomposable A -module has endomorphism ring k . The category $\mathcal{D}$ is Krull–Schmidt and directed. Its indecomposables are the images of the indecomposable A -modules with $d_X > 0$; they remain distinct and have endomorphism ring k .*
- (3) *For every object X ,*

$$\mathcal{D}(P, X) = \text{Hom}_A(P, X) \cong Xe, \quad \mathcal{D}(X, I) = \text{Hom}_A(X, I) \cong D_k(Xe). \quad (3.1)$$

The object P is the unique source and I the unique sink of the irreducible quiver of $\mathcal{D}$.

Proof. (1) If $f_e = 0$, then f kills XeA and factors through the B -module X/XeA . Conversely, a factorization through a B -module has zero evaluation. This proves faithfulness.

(2) Directedness gives $\text{End}_A(X) = k$ for every indecomposable X : a nonzero noninvertible endomorphism would be a cycle, and a finite-dimensional division algebra over the algebraically closed field k is k . For $d_X > 0$, the ideal of maps through B is zero in this ring, since a factorization of 1_X through a B -module would make X a direct summand of that module and force $Xe = 0$. Hence X remains indecomposable. If $f : X \rightarrow Y$ and $g : Y \rightarrow X$ lift inverse isomorphisms in the quotient, then $gf - 1_X$ and $fg - 1_Y$ belong to these zero ideals. Thus f, g are inverse over A , and nonisomorphic indecomposable A -modules with nonzero evaluation at e remain nonisomorphic in $\mathcal{D}$. Lifting a cycle of nonzero nonisomorphisms would give one over A , so the quotient is directed.

Every object of $\mathcal{D}$ is a finite sum of indecomposables with endomorphism ring k , so $\mathcal{D}$ is Krull–Schmidt.

(3) The identifications in (3.1) hold because $\text{Hom}_A(P, \text{mod } B) = 0$ and $\text{Hom}_A(\text{mod } B, I) = 0$. Every indecomposable of $\mathcal{D}$ therefore receives a nonzero map from P and maps nontrivially to I . An incoming arrow at P , or an outgoing arrow at I , would give a cycle. Expanding the nonzero maps from P and to I into paths by Lemma 2.12 gives an incoming arrow at every $X \neq P$ and an outgoing arrow at every $X \neq I$, by taking the last and first arrows of these paths, respectively. $\square$

We next identify the arrows that contribute to the quotient’s arrow count.

Lemma 3.4. *Let X, Y be indecomposable A -modules with $Xe \neq 0$ and $Ye \neq 0$. The quotient functor induces an isomorphism*

$$\text{Irr}_A(X, Y) \cong \text{Irr}_{\mathcal{D}}(X, Y).$$

Proof. Both factors of a map through an indecomposable B -module are radical, so $[\text{mod } B](X, Y) \subseteq \text{rad}_A^2(X, Y)$. If $X \not\cong Y$, their images remain nonisomorphic by Lemma 3.3(2), so every map between them is radical in both categories. If $X \cong Y$, both radical spaces are zero, since $\text{End}_A(X) = \text{End}_{\mathcal{D}}(X) = k$. Thus the quotient map sends $\text{rad}_A(X, Y)$ onto $\text{rad}_{\mathcal{D}}(X, Y)$, with kernel $[\text{mod } B](X, Y) \subseteq \text{rad}_A^2(X, Y)$. A composite of radical maps through an indecomposable with nonzero evaluation at e lifts to such a composite over A ; a composite through a B -module becomes zero. The image of $\text{rad}_A^2(X, Y)$ is therefore $\text{rad}_{\mathcal{D}}^2(X, Y)$, so the induced map on radical quotients is both surjective and injective. $\square$

3.2. The inherited τ -sequences. By Lemma 3.4, the quotient preserves irreducible maps between indecomposables with nonzero evaluation at e . Their Auslander–Reiten translates over A , however, may belong to $\text{mod } B$ and hence become zero in $\mathcal{D}$. The following proposition describes the resulting τ -category structure; bars denote images under $\text{mod } A \rightarrow \mathcal{D}$.

Proposition 3.5 ([9, Section 1.4(2),(3)]). *The category $\mathcal{D}$ is a τ -category. For an indecomposable A -module X with $Xe \neq 0$, its right and left τ -sequences are*

$$\overline{\tau_A X} \longrightarrow \overline{\theta_A X} \longrightarrow X, \quad X \longrightarrow \overline{\theta_A^- X} \longrightarrow \overline{\tau_A^- X},$$

using the projective and injective conventions of Definition 2.5. Thus each term is obtained by deleting its indecomposable summands in $\text{mod } B$.

Proof. Iyama’s result [9, Section 1.4(2),(3)] states that the quotient of a τ -category by a full additive subcategory closed under direct summands is again a τ -category. At a surviving indecomposable, the right τ -sequence is the image of the original one when the image of its middle term is nonzero; otherwise it is $0 \rightarrow 0 \rightarrow X$. The dual statement gives the left τ -sequences. Example 2.11 applies to $\text{mod } A$, and $\text{mod } B$ is a full additive subcategory closed under direct summands. It remains to check that the displayed images also have the stated form when a middle term vanishes. Indeed, if X is nonprojective over A , applying $(-)_e$ to its almost split sequence over A gives $d_{\theta_A X} = d_{\tau_A X} + d_X > 0$. Its middle term is therefore nonzero in the quotient. If the middle term becomes zero, X must therefore be projective over A . Then $\tau_A X = 0$, and the displayed image sequence is exactly $0 \rightarrow 0 \rightarrow X$, as required. The same argument applies to left τ -sequences by duality. $\square$

We now fix notation for the inherited terms and the objects with zero translate.

Definition 3.6. For $X \in \text{ind } \mathcal{D}$, write the inherited sequences as

$$\tau^+ X \longrightarrow \theta^+ X \longrightarrow X, \quad X \longrightarrow \theta^- X \longrightarrow \tau^- X.$$

Here $\tau^+ X$ is the image of $\tau_A X$, and similarly on the left; the sequences at projective and injective A -modules are included. Following the notation of [8, Section 2.1(3)], put

$$\text{ind}_1^+ \mathcal{D} = \{X : \tau^+ X = 0\}, \quad \text{ind}_1^- \mathcal{D} = \{X : \tau^- X = 0\}.$$

Their members are called τ -projective and τ -injective, respectively. Their union is the boundary. Write $\Gamma_{\mathcal{D}}$ for the irreducible quiver of $\mathcal{D}$ equipped with this translation, displayed by dotted arrows $X \dashrightarrow \tau^+ X$.

Remark 3.7. Let X be an indecomposable A -module with $Xe \neq 0$. Its image is τ -projective in $\mathcal{D}$ precisely when X is projective over A or $\tau_A X$ belongs to $\text{mod } B$. In particular, an object can be τ -projective in $\mathcal{D}$ without being projective over A . The dual statement holds for τ -injectives.

The inherited sequences give the following identities, which we will use in the arrow count and the multiplicity bounds.

Lemma 3.8. *In Setting 3.1, the following hold.*

(1) *The boundary sets have the same cardinality:*

$$|\text{ind}_1^+ \mathcal{D}| = |\text{ind}_1^- \mathcal{D}|. \quad (3.2)$$

(2) *For $X, Y \in \text{ind } \mathcal{D}$, the middle multiplicities are*

$$[\theta^+ X : Y]_{\oplus} = \dim_k \text{Irr}_{\mathcal{D}}(Y, X), \quad [\theta^- X : Y]_{\oplus} = \dim_k \text{Irr}_{\mathcal{D}}(X, Y). \quad (3.3)$$

(3) *If $\tau^+ X \neq 0$, then for every $Y \in \text{ind } \mathcal{D}$,*

$$\theta^-(\tau^+ X) \cong \theta^+ X, \quad \dim_k \text{Irr}_{\mathcal{D}}(Y, X) = \dim_k \text{Irr}_{\mathcal{D}}(\tau^+ X, Y). \quad (3.4)$$

(4) *For $X \in \text{ind } \mathcal{D}$, evaluation at e satisfies*

$$d_X - d_{\theta^+ X} + d_{\tau^+ X} = \mathbf{1}_{\{X=P\}}, \quad d_X - d_{\theta^- X} + d_{\tau^- X} = \mathbf{1}_{\{X=I\}}. \quad (3.5)$$

Here $\mathbf{1}_{\{X=P\}}$ is one if $X \cong P$ and zero otherwise, and similarly for I .

Proof. (1) Remark 2.10 gives inverse bijections

$$\tau^+ : \text{ind } \mathcal{D} \setminus \text{ind}_1^+ \mathcal{D} \longleftrightarrow \text{ind } \mathcal{D} \setminus \text{ind}_1^- \mathcal{D} : \tau^-.$$

Since $\text{ind } \mathcal{D}$ is finite, their complements have equal cardinality.

(2) Proposition 3.5 deletes only the B -summands from the middle terms. Lemma 3.4 identifies the remaining irreducible spaces with those in (2.2), proving (3.3).

(3) By Remark 2.10, the right τ -sequence at X is also the left τ -sequence at $\tau^+ X$. Their middle terms agree, and (2) gives the equality of arrow multiplicities.

(4) Deleted B -summands have zero evaluation at e , so they do not change d for any term. For a nonprojective A -module, the first identity follows by applying the exact functor $(-)_e$ to its almost split sequence. At a projective it is $d_X - d_{\text{rad } X} = [\text{top } X : E]$. An indecomposable projective has simple top, so this multiplicity is one precisely when $X \cong P$, and zero otherwise. The second identity is dual. $\square$

3.3. Multiplicity one. The multiplicity bounds below are consequences of Ringel–Vossieck’s results on finite hammocks. In their application to directed algebras, these include the estimate of V. Höhne recorded in [11, Section 10, p. 243]. Besides determining the arrows at the boundary of $\mathcal{D}$, the bounds control the endpoints of the almost split sequences compared in Section 4.

Proposition 3.9 ([11, Section 1, Corollary 3; Section 10, p. 243]). *In Setting 3.1, the following hold.*

- (1) *Every τ -projective and τ -injective of $\mathcal{D}$ has $d_X = 1$.*
- (2) *Every indecomposable projective or injective A -module has $d_X \leq 1$.*
- (3) *For an almost split sequence $0 \rightarrow X \rightarrow V \rightarrow Y \rightarrow 0$ over A ,*

$$Xe = 0 \implies d_Y \leq 1, \quad Ye = 0 \implies d_X \leq 1. \quad (3.6)$$

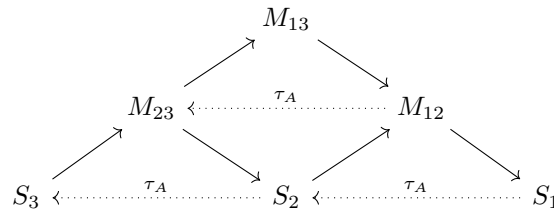
A direct proof is given in Appendix A.1.

Corollary 3.10. *In $\Gamma_{\mathcal{D}}$, each τ -projective other than P has exactly one incoming arrow, with multiplicity one; P has none. Dually, each τ -injective other than I has exactly one outgoing arrow, and I has none.*

Proof. For a τ -projective $X \neq P$, equation (3.5) gives $d_{\theta^+ X} = d_X = 1$. Every indecomposable summand of $\theta^+ X$ has positive integral dimension at e , so there is exactly one, occurring once. Equation (3.3) then gives the incoming arrow count. For P , Lemma 3.3(2),(3) gives $d_P = \dim_k \text{End}_A(P) = 1$, so equation (3.5) gives $d_{\theta^+ P} = 0$. Thus there are no incoming arrows. The second formula in (3.5) gives the outgoing counts by the same argument. $\square$

The following example shows how deleting modules changes the translation and produces additional τ -projective objects.

Example 3.11. Let Q be $1 \rightarrow 2 \rightarrow 3$ and $A = (kQ)^{\text{op}}$. Thus right A -modules are representations in the displayed direction. Write M_{ij} for the interval representation on vertices $i, \dots, j$, with identity maps between nonzero coordinates, and $S_i = M_{ii}$. The Auslander–Reiten translation quiver of A is



Take e to be the vertex idempotent at 2, so $E = S_2$. Then $B \cong k \times k$, whose indecomposables S_1, S_3 become zero in $\mathcal{D}$. The quotient has translation quiver

$$\begin{array}{ccccc} & & M_{13} & & \\ & \nearrow & & \searrow & \\ P = M_{23} & \xleftarrow{\tau^+} & & & M_{12} = I \\ & \searrow & & \nearrow & \\ & & S_2 & & \end{array}$$

The only nonzero translate is $\tau^+ M_{12} = M_{23}$. In particular, the right τ -sequence at S_2 has first term zero because its first term over A is S_3 . Hence $\text{ind}_1^+ \mathcal{D} = \{M_{23}, M_{13}, S_2\}$. Each has dimension one at vertex 2, as does $I = M_{12}$.

3.4. Path lengths and the number of arrows. The boundary arrows determine the lengths of paths in $\Gamma_{\mathcal{D}}$. We use these lengths to prove the bound $a(\mathcal{D}) \leq 2|\text{ind } \mathcal{D}| - |\text{ind}_1^+ \mathcal{D}| - 1$ needed for deletion.

Definition 3.12. Let $a(\mathcal{D}) = \sum_{X, Y \in \text{ind } \mathcal{D}} \dim_k \text{Irr}_{\mathcal{D}}(X, Y)$, and put

$$\varepsilon(\mathcal{D}) = 2|\text{ind } \mathcal{D}| - a(\mathcal{D}) - |\text{ind}_1^+ \mathcal{D}| - 1.$$

Thus the desired arrow bound is equivalent to $\varepsilon(\mathcal{D}) \geq 0$.

The quiver $\Gamma_{\mathcal{D}}$ is finite with unique source P , and each τ -projective has at most one incoming arrow, by Lemma 3.3(3) and Corollary 3.10. The lemma of Ringel–Vossieck [11, Section 7, pp. 234–235] gives the grading property: all paths of irreducible maps from P to a fixed indecomposable have the same length. Together with Lemma 2.12, this gives the height function and radical identities below.

Proposition 3.13. *There is a unique function $h : \text{ind } \mathcal{D} \rightarrow \mathbb{Z}_{\geq 0}$, called the height, with $h(P) = 0$ such that every arrow $X \rightarrow Y$ satisfies $h(Y) = h(X) + 1$, and $h(X) = h(\tau^+ X) + 2$ whenever $\tau^+ X \neq 0$. For distinct X, Y with $\mathcal{D}(X, Y) \neq 0$, put $\ell = h(Y) - h(X)$. Then $\ell > 0$ and*

$$\mathcal{D}(X, Y) = \text{rad}_{\mathcal{D}}^{\ell}(X, Y), \quad \text{rad}_{\mathcal{D}}^{\ell+1}(X, Y) = 0. \quad (3.7)$$

Put $L = h(I)$. All heights lie in $[0, L]$, with P, I the unique objects of heights 0, L , respectively.

A direct proof is given in Appendix A.2.

The next count reduces the arrow bound to $L \geq |\text{ind}_1^+ \mathcal{D}| - 1$, where $L = h(I)$.

Lemma 3.14. *With $L = h(I)$, one has*

$$a(\mathcal{D}) = 2|\text{ind } \mathcal{D}| - L - 2, \quad \varepsilon(\mathcal{D}) = L - (|\text{ind}_1^+ \mathcal{D}| - 1). \quad (3.8)$$

Proof. Put $H_+ = \sum_{X \in \text{ind}_1^+ \mathcal{D}} h(X)$ and $H_- = \sum_{X \in \text{ind}_1^- \mathcal{D}} h(X)$. The map τ^+ gives a bijection from the non- τ -projectives to the non- τ -injectives. Hence

$$\begin{aligned} \sum_{X \notin \text{ind}_1^+ \mathcal{D}} h(X) &= \sum_{X \in \text{ind } \mathcal{D}} h(X) - H_+, \\ \sum_{X \notin \text{ind}_1^+ \mathcal{D}} h(\tau^+ X) &= \sum_{Y \notin \text{ind}_1^- \mathcal{D}} h(Y) = \sum_{Y \in \text{ind } \mathcal{D}} h(Y) - H_-. \end{aligned}$$

Summing $h(X) - h(\tau^+ X) = 2$ over the non- τ -projectives gives

$$H_- - H_+ = 2(|\text{ind } \mathcal{D}| - |\text{ind}_1^+ \mathcal{D}|).$$

Every arrow contributes the height of its target minus that of its source, which is one. Thus

$$a(\mathcal{D}) = \sum_X h(X) \text{indeg}_{\mathcal{D}}(X) - \sum_X h(X) \text{outdeg}_{\mathcal{D}}(X).$$

For a non- τ -projective X , equation (3.4) gives $\text{indeg}_{\mathcal{D}}(X) = \text{outdeg}_{\mathcal{D}}(\tau^+ X)$. Reindexing by $Y = \tau^+ X$ and using Corollary 3.10 gives

$$\sum_X h(X) \text{indeg}_{\mathcal{D}}(X) = \sum_{Y \notin \text{ind}_1^- \mathcal{D}} (h(Y) + 2) \text{outdeg}_{\mathcal{D}}(Y) + H_+.$$

Here the boundary contribution is H_+ , since every τ -projective except P has indegree one and $h(P) = 0$. Similarly, every τ -injective except I has outdegree one, so

$$\sum_{Y \in \text{ind}_1^- \mathcal{D}} h(Y) \text{outdeg}_{\mathcal{D}}(Y) = H_- - L, \quad \sum_{Y \in \text{ind}_1^- \mathcal{D}} \text{outdeg}_{\mathcal{D}}(Y) = |\text{ind}_1^- \mathcal{D}| - 1.$$

Subtracting the outgoing weighted sum, we obtain

$$\begin{aligned} a(\mathcal{D}) &= 2 \sum_{Y \notin \text{ind}_1^- \mathcal{D}} \text{outdeg}_{\mathcal{D}}(Y) + H_+ - H_- + L \\ &= 2(a(\mathcal{D}) - |\text{ind}_1^+ \mathcal{D}| + 1) - 2(|\text{ind} \mathcal{D}| - |\text{ind}_1^+ \mathcal{D}|) + L. \end{aligned}$$

In the second equality, we subtract the boundary outdegree from the total outdegree $a(\mathcal{D})$, then use (3.2) and the formula for $H_- - H_+$. Solving gives the first formula; substitution in Definition 3.12 gives the second. The same calculation includes $P = I$, when there is one object and no arrows. $\square$

3.5. The poset and its representations. To bound L , we use Ringel–Vossieck’s identification of $\mathcal{D}$ with the category of representations by subspaces of a poset with one element for each τ -projective other than P . We give the poset and the functor explicitly.

Definition 3.15. For a finite poset T , a *representation of T by subspaces*, or T -space, is a finite-dimensional vector space W with subspaces W_t such that $W_s \subseteq W_t$ whenever $s \leq t$. A morphism $(W; W_t) \rightarrow (W'; W'_t)$ is a linear map $f : W \rightarrow W'$ such that $f(W_t) \subseteq W'_t$ for every $t \in T$. Write $T\text{-Sp}$ for this category. A subset $U \subseteq T$ is *upward-closed* if $s \in U$ and $s \leq t$ imply $t \in U$. For such U , let K_U have total space k , with subspace $(K_U)_t = k$ for $t \in U$ and $(K_U)_t = 0$ otherwise.

Construction 3.16. Index the τ -projectives other than P by a set T , choosing indecomposable A -module representatives Q_t . Define

$$s \leq t \iff \mathcal{D}(Q_t, Q_s) \neq 0.$$

Since $eAe \cong \text{End}_A(P) = k$ by Lemma 3.3(2),(3), identify $F(P) = eAe$ with k , using e as basis. By Proposition 3.9(1), each $F(Q_t)$ is one-dimensional. Choose a basis of each $F(Q_t)$, and normalize $u_t : P \rightarrow Q_t$ so that its evaluation sends 1 to that basis vector. Identify $\mathcal{D}(P, X)$ with $F(X) = Xe$ by (3.1), and set

$$F_t(X) = \text{Im}(\mathcal{D}(Q_t, X) \longrightarrow \mathcal{D}(P, X), f \longmapsto fu_t), \quad G(X) = (F(X); F_t(X)). \quad (3.9)$$

On morphisms, G is evaluation at e . Rescaling u_t does not change its Hom image $F_t(X)$. The chosen bases give coordinates for the module construction in Appendix A.3.

Proposition 3.17 ([11, Section 10, Theorem, p. 243]). *Construction 3.16 defines a finite poset and a k -linear equivalence $G : \mathcal{D} \xrightarrow{\sim} T\text{-Sp}$. Moreover,*

$$G(P) = K_{\emptyset}, \quad G(Q_t) = K_{\{s : t \leq s\}}, \quad G(I) = K_T, \quad |T| = |\text{ind}_1^+ \mathcal{D}| - 1.$$

A direct proof is given in Appendix A.3.

We compute the equivalence on the four objects of Example 3.11.

Example 3.18. For Example 3.11, let $Q_{t_1} = M_{13}$ and $Q_{t_2} = S_2$. They have no nonzero maps between them in $\mathcal{D}$, so T is a two-element antichain. Evaluation and the two indexed subspaces give

X	$F(X)$	$F_{t_1}(X)$	$F_{t_2}(X)$	$G(X)$
M_{23}	k	0	0	$K_{\emptyset}$
M_{13}	k	k	0	$K_{\{t_1\}}$
S_2	k	0	k	$K_{\{t_2\}}$
M_{12}	k	k	k	K_T

Each arrow of the displayed quotient quiver is the identity on total spaces. Both composites from $K_\emptyset$ to K_T are therefore the identity of k , so the four arrows form a commutative square. The heights are 0, 1, 1, 2, giving $L = 2 = |T|$ and $\varepsilon(\mathcal{D}) = 0$.

3.6. The arrow bound and its equality case. By Lemma 3.14 and Proposition 3.17, we have $\varepsilon(\mathcal{D}) = L - |T|$. Ringel–Vossieck prove that $L \geq |T|$ and that, if equality holds, every indecomposable T -space has one-dimensional total space [11, Section 8, Proposition, p. 237]. Since the total space of $G(X)$ is Xe , this gives the following inequality and its equality consequence.

Proposition 3.19. *One has $\varepsilon(\mathcal{D}) \geq 0$. If $\varepsilon(\mathcal{D}) = 0$, then $d_X = 1$ for every $X \in \text{ind } \mathcal{D}$.*

A direct proof is given in Appendix A.4.

The following example illustrates strictness of the arrow bound. Its poset is a three-element antichain, with $\varepsilon(\mathcal{D}) = 1$ and an indecomposable X with $\dim_k Xe = 2$.

Example 3.20. Let Q be the Dynkin quiver of type D_4 with arrows $1 \rightarrow 0$, $2 \rightarrow 0$, $3 \rightarrow 0$, and put $A = (kQ)^{\text{op}}$. Take e at vertex 0. The three τ -projectives other than $P = S_0$ are the projectives supported on $\{0, i\}$, so T is a three-element antichain. Its representations are triples of subspaces of a vector space. Besides the eight one-dimensional representations K_U , for $U \subseteq T$, there is an indecomposable Z with total space k^2 and indexed lines $k(1, 0)$, $k(0, 1)$, and $k(1, 1)$. As an A -module, K_U has space k at 0 and at the vertices in U , with identity maps between nonzero spaces. The module Z has space k^2 at 0, and its three arrow maps are the inclusions of the displayed lines. The quotient has nine indecomposables, twelve arrows, and four τ -projectives. Thus

$$\varepsilon(\mathcal{D}) = 18 - 12 - 4 - 1 = 1, \quad L = 4 > |T| = 3.$$

Example A.2 gives the full decomposition, Hom spaces, and translation quiver, and exhibits the extra step in path length.

4. DELETING A SIMPLE IN THE DIRECTED CASE

We retain Setting 3.1, with $B = A/AeA$. Proposition 3.19 bounds the number of arrows between indecomposables with $Xe \neq 0$. We now compare irreducible maps and almost split sequences over A and B to prove the deletion identity and inequality below. We first define the counts that occur in the identity.

Definition 4.1. For $X, Y \in \text{ind } B$, put

$$c_{XY} = \dim_k \text{Irr}_B(X, Y) - \dim_k \text{Irr}_A(X, Y), \quad c = \sum_{X, Y \in \text{ind } B} c_{XY}.$$

An almost split sequence over B is *new* if it is not almost split over A . Let r be the number of their isomorphism classes.

Theorem 4.2. *In Setting 3.1, with c, r as in Definition 4.1, one has*

$$\sigma(A) - \sigma(B) = \varepsilon(\mathcal{D}) + c - r \geq 0. \quad (4.1)$$

If equality holds in $\sigma(A) \geq \sigma(B)$, then $\dim_k Xe \leq 1$ for every indecomposable A -module X .

We prove the theorem in Subsection 4.3. Remark 2.4 extends its deletion inequality to representation-finite representation-directed algebras that are Morita equivalent to A . Since $\varepsilon(\mathcal{D}) \geq 0$ by Proposition 3.19, it remains to prove $c \geq r$. We start by interpreting c_{XY} as the dimension of a kernel between irreducible spaces.

Lemma 4.3. *For $X, Y \in \text{ind } B$, the identity on $\text{rad}_B(X, Y) = \text{rad}_A(X, Y)$ induces the surjection*

$$\text{Irr}_B(X, Y) \longrightarrow \text{Irr}_A(X, Y), \quad f + \text{rad}_B^2(X, Y) \longmapsto f + \text{rad}_A^2(X, Y).$$

Its kernel has dimension $c_{XY} \geq 0$.

Proof. The inclusion $\text{mod } B \hookrightarrow \text{mod } A$ is full, so $\text{rad}_B(X, Y) = \text{rad}_A(X, Y)$, while $\text{rad}_B^2(X, Y) \subseteq \text{rad}_A^2(X, Y)$: a factorization through a B -module is also one through an A -module. The kernel is $\text{rad}_A^2(X, Y) / \text{rad}_B^2(X, Y)$, whose dimension is the stated difference. $\square$

We may choose a basis of each kernel and extend it to a basis of $\text{Irr}_B(X, Y)$. The kernel basis elements will be called new arrows.

4.1. Restricting almost split sequences to the quotient algebra. To count the new almost split sequences, we recover the sequences over B using the largest B -submodule and largest B -quotient.

Definition 4.4. For $Z \in \text{mod } A$, put

$$\mathbf{R}_B Z = \{z \in Z : zJ = 0\}, \quad \mathbf{L}_B Z = Z/ZJ.$$

The adjunctions and vanishing properties below allow us to recover almost split sequences over B from those over A .

Lemma 4.5. (1) *The functors $\mathbf{R}_B$ and $\mathbf{L}_B$ are respectively right and left adjoint to $\text{mod } B \hookrightarrow \text{mod } A$. For every $Z \in \text{mod } A$, the quotient $Z/\mathbf{R}_B Z$ has no nonzero B -submodule, and the kernel of $Z \rightarrow \mathbf{L}_B Z$ has no nonzero B -quotient.*
(2) *A nonprojective indecomposable B -module Y is nonprojective over A . Write its almost split sequence over A as $0 \rightarrow \tau_A Y \rightarrow V \rightarrow Y \rightarrow 0$. Its almost split sequence over B is*

$$0 \longrightarrow \mathbf{R}_B(\tau_A Y) \longrightarrow \mathbf{R}_B V \longrightarrow Y \longrightarrow 0. \quad (4.2)$$

Dually, applying $\mathbf{L}_B$ to the almost split sequence over A starting at a noninjective B -module gives its almost split sequence over B .

Proof. (1) Every map from a B -module to Z has image in $\mathbf{R}_B Z$, and every map from Z to a B -module kills ZJ . These factorizations are unique, giving the adjunctions. If $Z/\mathbf{R}_B Z$ had a nonzero B -submodule, its inverse image would be a larger B -submodule of Z by the extension closure in Lemma 3.2. This is impossible. Dually, the kernel of $Z \rightarrow \mathbf{L}_B Z$ has no nonzero B -quotient.

(2) An A -projective annihilated by J is also B -projective, so Y is nonprojective over A . Every non-retraction $W \rightarrow Y$ with $W \in \text{mod } B$ factors through $V \rightarrow Y$, and its map into V has image in $\mathbf{R}_B V$. Thus the restriction is right almost split. It is nonsplit, since a section composed with $\mathbf{R}_B V \hookrightarrow V$ would split $V \rightarrow Y$. Since Y is not B -projective, its projective cover is not a retraction and factors through the restricted map. Hence that map is epic. Left exactness identifies its nonzero kernel with $\mathbf{R}_B(\tau_A Y)$.

Put $Q = (\tau_A Y)/\mathbf{R}_B(\tau_A Y)$. Since Q has no nonzero B -submodule, $\text{Hom}_A(Y, Q) = 0$. Auslander-Reiten duality (2.3) and $\tau_A^-(\tau_A Y) = Y$ give $\text{Ext}_A^1(Q, \tau_A Y) = 0$. Applying $\text{Hom}_A(-, \tau_A Y)$ to the sequence defining Q therefore gives a surjection

$$\text{End}_A(\tau_A Y) \longrightarrow \text{Hom}_A(\mathbf{R}_B(\tau_A Y), \tau_A Y) = \text{End}_A(\mathbf{R}_B(\tau_A Y)).$$

The last equality holds because every image lies in the largest B -submodule. By Lemma 3.3(2), this is a unital ring map from k to a nonzero ring. Hence $\text{End}_A(\mathbf{R}_B(\tau_A Y)) = k$, so the module $\mathbf{R}_B(\tau_A Y)$ is indecomposable. If the restricted map were not right minimal, we could write it as $(g', 0) : V' \oplus V'' \rightarrow Y$, with $V'' \neq 0$. The map g' is still a nonsplit epimorphism, so $\ker g' \neq 0$. The kernel of the restricted map would be $\ker g' \oplus V''$, contradicting the indecomposability of $\mathbf{R}_B(\tau_A Y)$. Thus (4.2) is almost split. Duality proves the assertion for $\mathbf{L}_B$. $\square$

To compare c and r , we determine the endpoints and middle summands of a new sequence.

Lemma 4.6. *Let $0 \rightarrow X_0 \rightarrow U \rightarrow Y_0 \rightarrow 0$ be a new almost split sequence in $\text{mod } B$. Then the following hold.*

(1) *Its endpoints satisfy*

$$\mathbf{R}_B(\tau_A Y_0) = X_0, \quad \mathbf{L}_B(\tau_A^- X_0) = Y_0, \quad (4.3)$$

and $d_{\tau_A Y_0} = d_{\tau_A^- X_0} = 1$.

(2) *The middle term V of the almost split sequence over A ending at Y_0 has a decomposition $V = V_0 \oplus Z$, with $V_0 \in \text{mod } B$, Z indecomposable, and $d_Z = 1$. Moreover,*

$$U = V_0 \oplus \mathbf{R}_B Z, \quad [\mathbf{R}_B Z : X]_{\oplus} = c_{X, Y_0} \quad (X \in \text{ind } B). \quad (4.4)$$

The dual assertion holds for the sequence over A starting at X_0 .

Proof. (1) Lemma 4.5(2) and its dual give (4.3). Both $\tau_A Y_0$ and $\tau_A^- X_0$ have nonzero evaluation at e . Otherwise the corresponding almost split sequence over A would lie in $\text{mod } B$ by Lemma 3.2, and its restriction would be the same sequence. This contradicts the assumption that the sequence is not almost split over A . Equation (3.6) then gives dimension one at e for both translates.

(2) Exact evaluation of the almost split sequence over A ending at Y_0 gives $d_V = d_{\tau_A Y_0} + d_{Y_0} = 1 + 0 = 1$. Thus exactly one indecomposable summand has nonzero evaluation, with multiplicity one. This gives $V = V_0 \oplus Z$ as stated. Lemma 4.5(2) gives $U = V_0 \oplus R_B Z$. For $X \in \text{ind } B$, we have $X \not\cong Z$, so (2.2) gives

$$[R_B Z : X]_{\oplus} = [U : X]_{\oplus} - [V_0 : X]_{\oplus} = \dim_k \text{Irr}_B(X, Y_0) - \dim_k \text{Irr}_A(X, Y_0).$$

This proves (4.4). The dual comparison at X_0 gives the multiplicities $c_{X_0, Y}$. $\square$

4.2. Ext and new almost split sequences. To prove $r \leq c$, we compare Ext dimensions at the endpoints and middle summands of new almost split sequences. The endpoints contribute one per sequence, while the middle summand multiplicities count new arrows. The Ext spaces involved are $\text{Ext}_A^1(E, X)$ and $\text{Ext}_A^1(X, E)$ for $X \in \text{ind } B$.

Definition 4.7. For $X \in \text{ind } B$, put

$$\eta_X = \dim_k \text{Ext}_A^1(E, X), \quad \zeta_X = \dim_k \text{Ext}_A^1(X, E).$$

Lemma 4.8. For $X, Y \in \text{ind } B$, the following hold.

(1) The Ext dimensions satisfy

$$\eta_X = \dim_k \text{Hom}_A(\tau_A^- X, E), \quad \zeta_X = \dim_k \text{Hom}_A(E, \tau_A X). \quad (4.5)$$

Both dimensions lie in $\{0, 1\}$.

(2) One has $\eta_X \zeta_X = 0$.

(3) An irreducible map gives the bound

$$\text{Irr}_B(X, Y) \neq 0 \implies \eta_X + \zeta_Y \leq 1. \quad (4.6)$$

Proof. (1) We first show that maps through projectives or injectives vanish in the two Hom spaces in Auslander–Reiten duality (2.3). If a nonzero map $\tau_A^- X \rightarrow E$ factors through projectives, some nonzero component factors through the projective cover P of E . Its map into P is nonzero on the top, hence epic by Nakayama’s lemma. It splits, so P is a direct summand of $\tau_A^- X$. Since $\tau_A^- X$ is indecomposable, it is projective, contradicting the nonprojectivity of a nonzero inverse Auslander–Reiten translate. If X is injective, both sides are zero. Dually, a nonzero map $E \rightarrow \tau_A X$ factoring through injectives has a nonzero factor through I . The map $I \rightarrow \tau_A X$ is injective on $\text{soc } I \cong E$, which is essential in I , hence monic and split, contradicting noninjectivity of $\tau_A X$. This proves (4.5). Since $Xe = 0$, (3.6) bounds the dimensions at e of the translates by one. For any module W , evaluation injects $\text{Hom}_A(W, E)$ into $\text{Hom}_k(We, Ee)$ and $\text{Hom}_A(E, W)$ into $\text{Hom}_k(Ee, We)$: a map vanishing at e has zero image in the first case and is zero on the generator of E in the second. Since $\dim_k Ee = 1$, both Hom dimensions are at most d_W . This gives the bounds on η_X, ζ_X .

(2) If $\text{Ext}_A^1(M, N) \neq 0$ for indecomposables, (2.3) gives a nonzero map $N \rightarrow \tau_A M$. Following it by the two maps through a middle summand of the almost split sequence ending at M gives a chain of nonzero maps from N to M . Omit an isomorphism step if it occurs; the remaining chain consists of nonisomorphisms. Nonzero Ext in both directions between E and X would therefore give a cycle. Thus $\eta_X \zeta_X = 0$.

(3) Choose an irreducible B -map $f : X \rightarrow Y$. Factoring it through its image shows that it is monic or epic. If it is monic, its cokernel is in $\text{mod } B$, so $\text{Hom}_A(E, \text{coker } f) = 0$. The long exact sequence gives an injection $\text{Ext}_A^1(E, X) \hookrightarrow \text{Ext}_A^1(E, Y)$. Hence $\eta_X = 1$ forces $\eta_Y = 1$, and then $\zeta_Y = 0$. If f is epic, $\text{Hom}_A(\ker f, E) = 0$ gives an injection $\text{Ext}_A^1(Y, E) \hookrightarrow \text{Ext}_A^1(X, E)$. Thus $\zeta_Y = 1$ forces $\zeta_X = 1$ and $\eta_X = 0$. In either case η_X and ζ_Y cannot both be one; part (1) therefore gives (4.6). $\square$

We now compute the total Ext dimension at the endpoints of a new sequence.

Lemma 4.9. *For a new almost split sequence $0 \rightarrow X_0 \rightarrow U \rightarrow Y_0 \rightarrow 0$,*

$$\eta_{X_0} + \zeta_{Y_0} = 1. \quad (4.7)$$

The inclusion $X_0 = R_B(\tau_A Y_0) \hookrightarrow \tau_A Y_0$ and quotient $\tau_A^- X_0 \twoheadrightarrow L_B(\tau_A^- X_0) = Y_0$ induce zero maps

$$\text{Ext}_A^1(E, X_0) \longrightarrow \text{Ext}_A^1(E, \tau_A Y_0), \quad \text{Ext}_A^1(Y_0, E) \longrightarrow \text{Ext}_A^1(\tau_A^- X_0, E). \quad (4.8)$$

Proof. First, for any almost split B -sequence, $\text{Hom}_A(E, Y_0) = 0$ gives an injection $\text{Ext}_A^1(E, X_0) \hookrightarrow \text{Ext}_A^1(E, U)$. If $\eta_{X_0} = 1$, some indecomposable middle summand W has $\eta_W = 1$. Its arrow to Y_0 and (4.6) imply $\zeta_{Y_0} = 0$. Thus the sum in (4.7) is at most one.

For a new sequence put $Q = (\tau_A Y_0)/X_0$. By (4.3) and Lemma 4.5(1), Q has no nonzero B -submodule. Lemma 4.6(1) gives $d_Q = 1$. Every simple in its socle must be E , and hence $\text{soc } Q \cong E$, so $\text{Hom}_A(E, Q) \cong k$. Applying $\text{Hom}_A(E, -)$ to $0 \rightarrow X_0 \rightarrow \tau_A Y_0 \rightarrow Q \rightarrow 0$ gives

$$0 \longrightarrow \text{Hom}_A(E, \tau_A Y_0) \longrightarrow k \longrightarrow \text{Ext}_A^1(E, X_0) \longrightarrow \text{Ext}_A^1(E, \tau_A Y_0).$$

By (4.5), $\dim_k \text{Hom}_A(E, \tau_A Y_0) = \zeta_{Y_0}$. The image of the connecting map from k has dimension $1 - \zeta_{Y_0}$, so $\eta_{X_0} \geq 1 - \zeta_{Y_0}$. This proves equality. It also proves the first vanishing in (4.8): either $\text{Ext}_A^1(E, X_0) = 0$, or the connecting map from k is onto this one-dimensional space. Applying this argument to the dual sequence proves the second vanishing. $\square$

Proposition 4.10. *The numbers in Definition 4.1 satisfy $r \leq c$.*

Proof. For a new sequence $0 \rightarrow X_0 \rightarrow U \rightarrow Y_0 \rightarrow 0$, use $U = V_0 \oplus R_B Z$ from (4.4). The injection $\text{Ext}_A^1(E, X_0) \hookrightarrow \text{Ext}_A^1(E, U)$ has zero component in $\text{Ext}_A^1(E, V_0)$. Indeed $X_0 \rightarrow U \rightarrow V_0$ is the restriction of $\tau_A Y_0 \rightarrow V_0 \oplus Z \rightarrow V_0$, so its Ext map factors through the first zero map in (4.8). Thus the injection takes values in $\text{Ext}_A^1(E, R_B Z)$. Equation (4.4) gives $R_B Z \cong \bigoplus_{X \in \text{ind } B} X^{c_{X, Y_0}}$, so additivity of Ext gives dimension $\sum_X c_{X, Y_0} \eta_X$. Comparing dimensions, and applying the dual argument to extensions of Y_0 by E , gives

$$\eta_{X_0} \leq \sum_X c_{X, Y_0} \eta_X, \quad \zeta_{Y_0} \leq \sum_Y c_{X_0, Y} \zeta_Y.$$

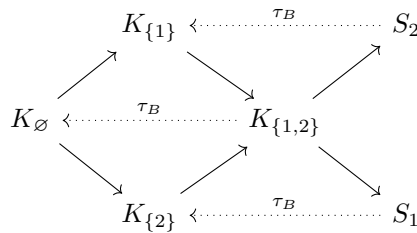
Let $\mathcal{N}$ be the set of pairs (X_0, Y_0) occurring as the endpoints of new sequences. Each right endpoint and each left endpoint occurs at most once, by AR translation over B . Summing the two inequalities and using (4.7) gives

$$\begin{aligned} r &\leq \sum_{(X_0, Y_0) \in \mathcal{N}} \sum_X c_{X, Y_0} \eta_X + \sum_{(X_0, Y_0) \in \mathcal{N}} \sum_Y c_{X_0, Y} \zeta_Y \\ &\leq \sum_{X, Y \in \text{ind } B} c_{XY} \eta_X + \sum_{X, Y \in \text{ind } B} c_{XY} \zeta_Y = \sum_{X, Y \in \text{ind } B} c_{XY} (\eta_X + \zeta_Y). \end{aligned}$$

The second inequality adds the omitted endpoints to sums with nonnegative terms. If $c_{XY} > 0$, then $\text{Irr}_B(X, Y) \neq 0$, so (4.6) gives $\eta_X + \zeta_Y \leq 1$. Therefore $r \leq \sum_{X, Y} c_{XY} = c$. $\square$

Deleting a leaf in the preceding D_4 example gives a strict inequality $r < c$. We compute the new arrows and sequences explicitly.

Example 4.11. Keep $A = (kQ)^{\text{op}}$ with arrows $i \rightarrow 0$, $i = 1, 2, 3$, from Example 3.20. Write e_i for the idempotent at vertex i , and now take $E = S_3$ and $B = A/Ae_3A$. Keep the notation K_U, Z from that example. The indecomposable B -modules are $K_\emptyset, K_{\{1\}}, K_{\{2\}}, K_{\{1,2\}}, S_1, S_2$. The Auslander–Reiten translation quiver Γ_B is



The two arrows out of $K_\emptyset$ are irreducible over A . The other four are new. Their factorizations in $\mathbf{mod} A$ are visible in the following diagram of maps:

$$\begin{array}{ccccc}
 & K_{\{1\}} & & & S_1 \\
 & \searrow & & & \nearrow \\
 & Z & \longrightarrow & K_{\{1,2\}} & \longrightarrow & K_{\{1,2,3\}} \\
 & \nearrow & & & \searrow \\
 & K_{\{2\}} & & & S_2
 \end{array}$$

Each displayed map is irreducible in $\mathbf{mod} A$. The two composites through Z and the two through $K_{\{1,2,3\}}$ are the four new irreducible maps of $\mathbf{mod} B$. Both intermediate modules have nonzero space at vertex 3. In the displayed Γ_B , each of these four pairs has one arrow, and Γ_A has none between that pair, so each contributes one to c . The other two arrows are irreducible over A , and there are no other arrows in Γ_B . Thus $c = 4$.

The three almost split sequences of $\mathbf{mod} B$ are

$$\begin{aligned}
 0 \rightarrow K_\emptyset \rightarrow K_{\{1\}} \oplus K_{\{2\}} \rightarrow K_{\{1,2\}} \rightarrow 0, \\
 0 \rightarrow K_{\{2\}} \rightarrow K_{\{1,2\}} \rightarrow S_1 \rightarrow 0, \quad 0 \rightarrow K_{\{1\}} \rightarrow K_{\{1,2\}} \rightarrow S_2 \rightarrow 0.
 \end{aligned}$$

None is almost split over A : the almost split sequences over A with the same last terms start at $K_{\{3\}}, K_{\{2,3\}}, K_{\{1,3\}}$, respectively. Hence $r = 3$, and $c - r = 1$.

Apply $\mathrm{Hom}_A(-, X)$ to the projective resolution of $E = S_3$ $0 \rightarrow S_0 \rightarrow K_{\{3\}} \rightarrow S_3 \rightarrow 0$, where $S_0 = K_\emptyset$. For $X \in \mathbf{ind} B$, $\mathrm{Hom}_A(K_{\{3\}}, X) \cong X e_3 = 0$ and $\mathrm{Hom}_A(S_0, X) \cong X e_0$, so $\eta_X = \dim_k X e_0$. Since 3 is a source of Q , S_3 is injective, and $\zeta_Y = 0$ for all Y . Writing each of the three displayed almost split sequences as $0 \rightarrow X_0 \rightarrow U \rightarrow Y_0 \rightarrow 0$, we have $\eta_{X_0} + \zeta_{Y_0} = 1$; in this example, each new arrow also has $\eta_X + \zeta_Y = 1$. The estimate $r \leq c$ reads $3 \leq 4$. The strict inequality occurs at the first displayed sequence: $\eta_{K_\emptyset} = 1$, whereas its two new incoming arrows give $\sum_X c_{X, K_{\{1,2\}}} \eta_X = 2$. Example 4.16 combines these counts with those of $\mathcal{D}$ to check (4.1).

4.3. The deletion identity and the directed case. We have proved $\varepsilon(\mathcal{D}) \geq 0$ in Proposition 3.19 and $c - r \geq 0$ in Proposition 4.10. To deduce $\sigma(A) \geq \sigma(B)$, it remains to prove the deletion identity (4.1).

Proof of Theorem 4.2. Let z be the number of arrows of Γ_A with one endpoint in $\mathbf{ind} B$ and the other having nonzero evaluation at e . Partition the arrows into those with both endpoints having nonzero evaluation, those counted by z , and those with both endpoints in $\mathbf{ind} B$. Lemma 3.4 counts the first class by $a(\mathcal{D})$, and Definition 4.1 counts the last by $a(B) - c$. Thus

$$a(A) = a(\mathcal{D}) + z + a(B) - c, \quad |\mathbf{ind} A| = |\mathbf{ind} \mathcal{D}| + |\mathbf{ind} B|, \quad |A| = |B| + 1.$$

Put $p = |\mathbf{ind}_1^+ \mathcal{D}|$. Applying the surplus formula in Lemma 2.2 to A and B and subtracting gives

$$\begin{aligned}
 \sigma(A) - \sigma(B) &= 2(|\mathbf{ind} \mathcal{D}| - 1) - a(\mathcal{D}) - z + c \\
 &= (2|\mathbf{ind} \mathcal{D}| - a(\mathcal{D}) - p - 1) + p - 1 + c - z.
 \end{aligned}$$

By Definition 3.12, this is

$$\sigma(A) - \sigma(B) = \varepsilon(\mathcal{D}) + p - 1 + c - z. \quad (4.9)$$

Almost split sequences are indexed by their nonprojective indecomposable last terms, so there are $|\mathbf{ind} A| - |A|$ over A and $|\mathbf{ind} B| - |B|$ over B . We count those over A by their endpoints. There are $|\mathbf{ind} \mathcal{D}| - p$ with both endpoints having nonzero evaluation at e , since these are precisely the pairs $(\tau^+ X, X)$ with $\tau^+ X \neq 0$ in $\mathcal{D}$. There are $|\mathbf{ind} B| - |B| - r$ with both endpoints in $\mathbf{mod} B$: Lemma 3.2 puts their middle terms in $\mathbf{mod} B$, and they are exactly the almost split B -sequences that are also almost split over A .

The remaining sequences have exactly one endpoint in $\mathbf{mod} B$. We show that they are in bijection with the z arrows just counted. Consider first an irreducible map $f : X \rightarrow Y$ with $X \in \mathbf{ind} B$ and $Y e \neq 0$. The map f is monic or epic. If X were injective, f could not be monic, since it would then split. Nor can f be epic, since its target would belong to $\mathbf{mod} B$. Thus X is not

injective. In the almost split sequence starting at X , the middle term contains Y . By (3.6), the other endpoint has e -dimension at most one; exact evaluation at e shows that it has dimension one and that the middle term has exactly one summand with nonzero evaluation at e , with multiplicity one. Thus $X \rightarrow Y$ is the unique arrow from X to a summand of this middle term with nonzero evaluation at e . Conversely, every almost split sequence with first term in $\text{mod } B$ and last term having nonzero evaluation at e has this form. The dual argument treats arrows $X \rightarrow Y$ with $Xe \neq 0$ and $Y \in \text{mod } B$: an irreducible map into a projective is monic. If Y were projective over A , this would embed X into a B -module and force $Xe = 0$, a contradiction. The two families are disjoint, according as the first or last endpoint belongs to $\text{mod } B$, and exhaust the sequences with exactly one such endpoint. Consequently,

$$|\text{ind } A| - |A| = (|\text{ind } \mathcal{D}| - p) + (|\text{ind } B| - |B| - r) + z,$$

which gives $r = z - p + 1$. Substitution in (4.9) proves the identity. Proposition 3.19 and Proposition 4.10 show that both terms on the right of (4.1) are nonnegative. Equality forces $\varepsilon(\mathcal{D}) = 0$, so Proposition 3.19 gives dimension one at e for every indecomposable with $Xe \neq 0$; the other indecomposables have dimension zero. $\square$

We recall the classes of algebras that enter the equality argument.

Definition 4.12. A basic algebra is *special biserial* if it has a presentation $kQ/\mathfrak{r}$, with $J_Q^m \subseteq \mathfrak{r} \subseteq J_Q^2$ for the arrow ideal J_Q and some $m \geq 2$, satisfying two conditions: at most two arrows enter and at most two leave each vertex; and for each arrow α , there is at most one arrow β for which $\beta\alpha$ is defined and does not belong to $\mathfrak{r}$, and at most one arrow γ with the same property for $\alpha\gamma$. A non-basic algebra is called special biserial if its basic algebra is special biserial.

Definition 4.13. A module is *uniserial* if its submodules are totally ordered by inclusion. An algebra is *biserial* if the radical of every indecomposable projective, on either side, is a sum of two uniserial submodules whose intersection is simple or zero. A module over a basic algebra is *thin* if each simple occurs in it with multiplicity at most one, equivalently if $\dim_k Xe \leq 1$ at every primitive idempotent in a complete orthogonal set.

Every basic representation-finite biserial algebra is special biserial by Theorem 5.16 below. We first obtain biseriality in the directed case from the following result.

Proposition 4.14 ([10, Proposition 1]). *If every indecomposable module over a basic finite-dimensional algebra is thin, then the algebra is representation-finite and biserial.*

Corollary 4.15. *For a representation-finite representation-directed algebra A , $\sigma(A) \geq 0$. If A is basic and $\sigma(A) = 0$, then every indecomposable A -module is thin and A is biserial.*

Proof. By Morita invariance we may assume that A is basic. Successive primitive deletions remain representation-finite and representation-directed and end at the zero algebra, whose surplus is zero. Theorem 4.2 therefore proves nonnegativity. If $\sigma(A) = 0$, delete any primitive idempotent e first. Since $\sigma(B) \geq 0$ for $B = A/AeA$, we have $\sigma(A) = \sigma(B) = 0$, so every indecomposable has e -dimension at most one. As e was arbitrary, all indecomposables are thin. Proposition 4.14 gives biseriality. $\square$

The two deletions in our D_4 example distinguish the two nonnegative terms in (4.1).

Example 4.16. The indecomposables and arrows in Example A.2(2) give

$$|\text{ind } A| = 12, \quad |A| = 4, \quad a(A) = 15, \quad \chi(\text{mod } A) = 5, \quad \sigma(A) = 1.$$

Deleting vertex 0 gives $B_0 \cong k^3$. Deleting vertex 3 gives the three-vertex path algebra B_3 in Example 4.11. The complete counts are

e	$ \text{ind } B $	$ B $	$a(B)$	$\varepsilon(\mathcal{D})$	c	r	$\sigma(A) - \sigma(B)$
e_0	3	3	0	1	0	0	1
e_3	6	3	6	0	4	3	1

For the second row, the associated translation quiver $\Gamma_{\mathcal{D}}$ is

$$\begin{array}{ccccc}
 & & K_{\{1,3\}} & & \\
 & \nearrow & & \searrow & \\
 K_{\{3\}} & \rightarrow & Z & \xleftarrow{\tau^+} & K_{\{1,2,3\}} \rightarrow S_3 \\
 & \searrow & & \nearrow & \\
 & & K_{\{2,3\}} & &
 \end{array}$$

Thus it has six indecomposables and six solid arrows. Only $K_{\{1,2,3\}}$ has a nonzero τ^+ -translate, namely Z . Hence there are five τ -projectives, and $\varepsilon(\mathcal{D}) = 12 - 6 - 5 - 1 = 0$. The strict increase comes entirely from $\varepsilon(\mathcal{D})$ in the first row and from $c - r$ in the second.

5. THE GENERAL CASE VIA GRADED MODULES

Theorem 4.2 applies to representation-directed algebras. To treat an arbitrary representation-finite algebra, we construct representation-finite representation-directed algebras from graded modules over its standard form. We first recall the standard-form theorem and define the grading. We then count modules supported in finite intervals of degrees, and use these counts for the inequality and the equality case.

5.1. Standard forms and graded indecomposables. We recall the categorical description of the standard form. For an Auslander–Reiten translation quiver, the k -linear path category has the quiver vertices as objects; its morphisms are linear combinations of oriented paths, including the stationary identity paths, and composition is concatenation. The translation is denoted by τ . A *mesh presentation* chooses, at each nonprojective vertex X , a pairing of the arrows $b_{Y,j} : \tau X \rightarrow Y$ and $a_{Y,j} : Y \rightarrow X$. The mesh relation is

$$\sum_{Y,j} a_{Y,j} b_{Y,j} = 0.$$

The *mesh category* is the path category modulo these relations. In its realization by modules over a standard form, the paired arrows are represented by components belonging to the same copies of Y in an almost split sequence

$$0 \longrightarrow \tau X \xrightarrow{(b_{Y,j})_{Y,j}} \bigoplus_{Y,j} Y \xrightarrow{(a_{Y,j})_{Y,j}} X \longrightarrow 0.$$

Theorem 5.1 ([3, Sections 2.4, 2.8, 2.9, and 5.1]; [5, Section 3.1]). *Let A be a basic representation-finite algebra. There is a basic representation-finite algebra S , called a standard form of A , with the following properties.*

- (1) *The full Auslander–Reiten translation quivers of A and S are isomorphic, preserving arrow multiplicities, translation, and projective and injective vertices.*
- (2) *The category of indecomposable S -modules is k -linearly equivalent to the mesh category of this quiver, identifying vertices with the corresponding modules.*

We use path length in a mesh presentation to grade the Hom spaces.

Construction 5.2. For a basic representation-finite algebra A , choose a standard form S , a mesh presentation, and an equivalence as in Theorem 5.1. For $X, Y \in \text{ind } S$, let $\text{Hom}^d(X, Y)$ be the image under this equivalence of the subspace spanned by paths of length d . Put $\text{Hom}^d(X, Y) = 0$ for $d < 0$.

The following lemma records the properties of these subspaces and the numerical comparison with A .

Lemma 5.3. (1) *The subspaces in Construction 5.2 give a grading*

$$\text{Hom}_S(X, Y) = \bigoplus_{d \geq 0} \text{Hom}^d(X, Y), \quad \text{Hom}^0(X, Y) = \begin{cases} k1_X & X = Y, \\ 0 & X \neq Y, \end{cases} \quad (5.1)$$

with composition adding degrees. The positive-degree part is the radical, and the components of degree at least two give its square. In particular, $\text{Hom}^1(X, Y) \cong \text{Irr}_S(X, Y)$.

(2) One has $|S| = |A|$ and

$$\sigma(S) = \sigma(A). \quad (5.2)$$

Proof. (1) The mesh relations are homogeneous of degree two, so the path spaces give the grading (5.1). The positive-degree part is a nilpotent ideal with semisimple quotient: the indecomposable category is finite and all its Hom spaces are finite-dimensional. It is therefore the radical. Every path of length at least two is a product of arrows, so those components give the radical square and the degree-one components give the irreducible spaces.

(2) Theorem 5.1(1) preserves the number of indecomposables, arrows, and indecomposable projectives. The last number is the number of simples, so $|S| = |A|$. Lemma 2.2 then gives (5.2). $\square$

We now use the graded Hom spaces from the projectives to construct gradings on the algebra and its modules. These gradings refer to the fixed mesh presentation and equivalence. Path length here is measured in the Auslander–Reiten quiver, not in the Gabriel quiver of the algebra.

Setting 5.4. Throughout Section 5, choose a standard form S of a basic representation-finite algebra A , and fix the grading (5.1). Let $P_1, \dots, P_{|S|}$ be the indecomposable projectives, with $P_i = e_i S$. Their sum is the regular right module, and composition identifies

$$\text{End}_S\left(\bigoplus_i P_i\right) = S.$$

This gives S a nonnegative grading $S^{(d)} = \bigoplus_{i,j} \text{Hom}^d(P_i, P_j)$, with $S^{(0)} = \bigoplus_i k e_i$ and $\text{rad } S = \bigoplus_{d>0} S^{(d)}$. For $X \in \text{ind } S$, put

$$M_X = \text{Hom}_S\left(\bigoplus_i P_i, X\right), \quad (M_X)_d = \bigoplus_i \text{Hom}^d(P_i, X),$$

with right action by composition: after identifying $\bigoplus_i P_i = S$, $(f \cdot a)(u) = f(au)$ for $a, u \in S$. After forgetting the grading, M_X is X . Choose an integer $h \geq 1$ bounding all nonzero degrees in (5.1); such a bound exists because there are finitely many finite-dimensional Hom spaces. In particular $S^{(d)} = 0$ for $d > h$, and $(M_X)_d = 0$ for $d \notin [0, h]$.

Definition 5.5. A graded S -module is a decomposition $M = \bigoplus_{n \in \mathbb{Z}} M_n$ with $M_n S^{(d)} \subseteq M_{n+d}$. Its support is the set of integers n for which $M_n \neq 0$. Thus every M_X is supported in $[0, h]$. Write $\text{grmod } S$ for the category of finite-dimensional graded S -modules and degree-zero module maps, and Hom_{gr} for its morphism spaces. Write $M\langle s \rangle$ for the shift $(M\langle s \rangle)_r = M_{r-s}$. A homogeneous map of degree d sends M_r to M_{r+d} ; it becomes a map of degree zero from $M\langle s \rangle$ to $N\langle t \rangle$ when $d = s - t$.

We can now describe all graded indecomposables and the maps between them.

Proposition 5.6. *The category $\text{grmod } S$ has the following properties.*

(1) For $X, Y \in \text{ind } S$ and $s, t \in \mathbb{Z}$,

$$\text{Hom}_{\text{gr}}(M_X\langle s \rangle, M_Y\langle t \rangle) = \text{Hom}^{s-t}(X, Y). \quad (5.3)$$

(2) Every indecomposable is isomorphic to exactly one $M_X\langle s \rangle$, where $X \in \text{ind } S$ and $s \in \mathbb{Z}$.

(3) The category is directed.

(4) Its irreducible maps are the nonzero maps $M_X\langle t+1 \rangle \rightarrow M_Y\langle t \rangle$ corresponding to $\text{Hom}^1(X, Y)$ under (5.3). These maps have degree zero between the shifted modules.

Proof. (1) Evaluation at 1 identifies $\text{Hom}_S(S, X)$ with X . Under this identification a module map $M_X \rightarrow M_Y$ is postcomposition by a unique map $X \rightarrow Y$. A degree- d component of the latter sends $\text{Hom}^n(P_i, X)$ into $\text{Hom}^{n+d}(P_i, Y)$, and hence induces a degree- d module map. Decomposing into homogeneous components shows that the identification preserves degrees. With the shift convention of Definition 5.5, this gives (5.3).

(2) Let V be an indecomposable graded module. The graded category is Hom-finite and idempotent complete, hence Krull–Schmidt. Since V is indecomposable, $\text{End}_{\text{gr}}(V)$ is local. Decompose the underlying ungraded module and choose maps

$$f_j : V \longrightarrow M_{X_j}, \quad g_j : M_{X_j} \longrightarrow V, \quad \sum_j g_j f_j = 1_V.$$

Every ungraded map between finite-dimensional graded modules is a finite sum of homogeneous module maps: compare homogeneous components in its compatibility with multiplication by a homogeneous element of S . Taking degree zero in the displayed identity gives

$$1_V = \sum_{j,d} (g_j)_{-d} (f_j)_d.$$

Some summand is invertible, since the nonunits of the local endomorphism ring form an ideal. The corresponding pair of homogeneous maps exhibits V as a graded direct summand of $M_{X_j} \langle -d \rangle$. That module is indecomposable even without the grading, so the summand is the whole module. The underlying module determines X , and its least nonzero degree then determines the shift, proving uniqueness.

(3) By (5.1) and (5.3), each indecomposable has graded endomorphism ring k , and a nonzero map between distinct indecomposables strictly decreases the shift. This proves directedness.

(4) A composite of two radical maps has degree at least two, so a nonzero degree-one map cannot belong to the radical square. Conversely, every map of degree at least two is a sum of paths in degree-one maps, and hence factors through the corresponding intermediate graded modules; it belongs to the square of the radical. $\square$

5.2. Finite interval algebras and the lower bound. For each $m \geq 0$, we construct an algebra S_m whose modules are the graded S -modules supported in degrees $0, \dots, m$. We will show that S_m is directed and that $\sigma(S_m) = (m+1)\sigma(S) + O(1)$. Since $\sigma(S_m) \geq 0$ by Corollary 4.15, dividing by $m+1$ and letting $m \rightarrow \infty$ will prove $\sigma(S) \geq 0$.

Construction 5.7. For $m \geq 0$, let S_m be the algebra of the finite k -category with objects (i, r) , where $1 \leq i \leq |S|$ and $0 \leq r \leq m$, and morphism spaces

$$\text{Hom}((i, r), (j, s)) = (e_j S e_i)_{r-s}. \quad (5.4)$$

Negative homogeneous components are zero, and composition is multiplication in S . The algebra of a finite category is the direct sum of its morphism spaces, with product zero for noncomposable pairs. Ordering objects by degree gives a triangular matrix algebra with $m+1$ diagonal blocks $S^{(0)} = k^{|S|}$. Its semisimple quotient is $k^{(m+1)|S|}$, so S_m is basic and

$$|S_m| = (m+1)|S|.$$

We call S_m the *interval algebra* for the degrees $0, \dots, m$.

Proposition 5.8. *The category $\text{mod } S_m$ is equivalent to the full subcategory of graded S -modules supported in $[0, m]$. Its indecomposables are precisely the shifts $M_X \langle t \rangle$ supported in that interval. Consequently, S_m is representation-finite and representation-directed.*

Proof. Right modules over the algebra of a finite k -category are contravariant k -linear functors from that category to $\text{mod } k$. A graded module V assigns the space $V_r e_i$ to (i, r) . An element $a \in (e_j S e_i)_{r-s}$ acts by right multiplication

$$V_s e_j \longrightarrow V_r e_i, \quad v \longmapsto va.$$

Conversely, these action maps give a graded S -module by putting zero outside $[0, m]$. Associativity also holds across the boundary: degrees can only increase under multiplication, so if the starting and final degrees of a product belong to $[0, m]$, every intermediate degree belongs to the same interval. These inverse constructions preserve maps, direct sums, and exact sequences. Proposition 5.6 classifies the indecomposables and proves directedness. There are only finitely many shifts with the prescribed support. $\square$

Proposition 5.9. *With S and h fixed, as $m \rightarrow \infty$ we have*

$$|\operatorname{ind} S_m| = (m+1)|\operatorname{ind} S| + O(1), \quad (5.5)$$

$$a(S_m) = (m+1)a(S) + O(1), \quad (5.6)$$

$$\sigma(S_m) = (m+1)\sigma(S) + O(1). \quad (5.7)$$

Here every $O(1)$ is bounded independently of m .

Proof. For each X , let $l_X, u_X \in [0, h]$ be the least and greatest nonzero degrees of M_X . The allowed shifts are

$$-l_X \leq t \leq m - u_X.$$

For $m \geq h$ their number is $m+1 - (u_X - l_X)$. Summing over $X \in \operatorname{ind} S$ proves (5.5).

We count arrows by their targets. Equation (5.3) implies that a nonzero map $M_X\langle s \rangle \rightarrow M_Y\langle t \rangle$ requires

$$t \leq s \leq t + h. \quad (5.8)$$

Suppose $m \geq 2h$ and $0 \leq t \leq m - 2h$. Every possible incoming indecomposable is then supported in $[t, t+2h] \subseteq [0, m]$. The same holds for every intermediate object in a nonzero composite of two radical maps, since that object too has a nonzero map to the target. All relevant Hom spaces and factorizations therefore occur in $\operatorname{mod} S_m$. By Proposition 5.6, the indegree of $M_Y\langle t \rangle$ is exactly $\operatorname{indeg}_S(Y)$.

All allowed shifts lie in $[-h, m]$. The remaining shifts lie in $[-h, 0)$ or $(m-2h, m]$, with at most $h+2h=3h$ values per Y . Thus there are at most $3h|\operatorname{ind} S|$ remaining targets. For each such target $M_Y\langle t \rangle$, its indegree in $\operatorname{mod} S_m$ is at most $\sum_{X \in \operatorname{ind} S} \dim_k \operatorname{Hom}_S(X, Y)$, by summing (5.3) over the possible sources. Restricting to the interval may make a map irreducible by removing objects through which it factors. The Hom bound includes these arrows, since every irreducible space is a quotient of a subspace of Hom. The bound is independent of m and of the target shift. The total indegree of the remaining targets is therefore $O(1)$, and the interior targets contribute $(m-2h+1)a(S)$. This proves (5.6). Finally, $|S_m| = (m+1)|S|$ and Lemma 2.2 give (5.7). $\square$

Theorem 5.10. *Every representation-finite algebra A satisfies $\chi(\operatorname{mod} A) \geq |A|$.*

Proof. By Lemma 2.3, we may assume that A is basic. Choose a standard form S . By Corollary 4.15 and Proposition 5.8, $\sigma(S_m) \geq 0$ for every m . Divide (5.7) by $m+1$ and let $m \rightarrow \infty$. This gives $\sigma(S) \geq 0$, and (5.2) gives $\sigma(A) \geq 0$. $\square$

For the dual numbers we can describe the grading and all the interval algebras explicitly.

Example 5.11. Let $S = k[t]/(t^2)$. Its indecomposables are $P = S$ and $L = S/(t)$, with irreducible maps the quotient $\pi : P \rightarrow L$ and the socle inclusion $\iota : L \rightarrow P$. The cycle $P \rightarrow L \rightarrow P$ shows that S is not representation-directed, and Lemma 2.2 gives $\sigma(S) = 2(2-1) - 2 = 0$. Every morphism is a linear combination of identities, π , ι , and $\iota\pi$, subject to the mesh relation $\pi\iota = 0$. Thus the indecomposable category is its mesh category, and S may be taken as its own standard form. Since $\iota\pi$ is multiplication by t , path length gives t degree two. Accordingly, M_P has one-dimensional components in degrees zero and two, while M_L is concentrated in degree one.

For $m \geq 2$, the interval algebra S_m has vertices $0, \dots, m$, arrows $r \rightarrow r-2$ for $2 \leq r \leq m$, and all paths of length two zero. Its indecomposables are the $m+1$ simples and the $m-1$ shifts of M_P supported on a pair of degrees differing by two. Each nonsimple indecomposable has an irreducible map from its socle and one to its top. Hence

$$|S_m| = m+1, \quad |\operatorname{ind} S_m| = 2m, \quad a(S_m) = 2(m-1), \quad \sigma(S_m) = 0.$$

This also illustrates why the algebra grading need not put its Gabriel quiver arrows in degree one: the grading comes from paths between all indecomposable modules, including the intermediate simple L .

5.3. Zero surplus for finite intervals. To study the case $\sigma(S) = 0$, we need to show that $\sigma(S_r) = 0$ for every r . We prove the stronger bound below by realizing arbitrarily many copies of S_r as a quotient of a larger interval algebra and applying Theorem 4.2.

Proposition 5.12. *For every $r \geq 0$,*

$$0 \leq \sigma(S_r) \leq (r + h + 1)\sigma(S).$$

In particular, if $\sigma(S) = 0$, then every S_r has surplus zero, every indecomposable S_r -module is thin, and S_r is biserial.

Proof. Fix r , put $\ell = r + h + 1$, and let $q \geq 1$. Inside the interval $[0, M]$, where $M = q\ell - h - 1$, retain the q intervals

$$[j\ell, j\ell + r], \quad 0 \leq j < q,$$

and delete the vertices in the gaps by quotienting out the ideal generated by their primitive idempotents. Each gap has h degrees, so different retained intervals have no morphisms between them, by $S^{(d)} = 0$ for $d > h$. A morphism within one retained interval cannot factor through a deleted vertex: degrees are monotone along nonzero morphisms, so every intermediate degree remains in that same interval. Let f be the sum of the primitive idempotents at the deleted vertices. The preceding observations give

$$S_M / S_M f S_M \cong \prod_{j=0}^{q-1} S_r.$$

Perform this quotient one deleted primitive idempotent at a time. Each intermediate algebra is basic, representation-finite and directed by Lemma 3.2, applied to that algebra and the next primitive idempotent. Successive applications of Theorem 4.2, together with additivity on products from Lemma 2.3, give

$$q\sigma(S_r) \leq \sigma(S_M) = (q\ell - h)\sigma(S) + O(1).$$

The error is bounded independently of q by Proposition 5.9. Dividing by q and letting $q \rightarrow \infty$ gives the upper bound. Nonnegativity and the assertions in the equality case follow from Corollary 4.15. $\square$

5.4. Special biseriality and the main theorem. If $\sigma(A) = 0$, then $\sigma(S) = 0$ by (5.2), so Proposition 5.12 shows that every S_m is biserial. We compare almost split sequences over S and S_{h+2} to prove that S is biserial, then transfer the resulting bound to A through their common Auslander-Reiten quiver. We use the following characterization of biserial algebras due to Skowroński and Waschbüsch [12].

Definition 5.13. For a representation-finite algebra C , let $\beta_R(C)$ be the maximum number of nonprojective indecomposable summands in the middle term of an almost split sequence of right C -modules, counting multiplicities. An empty maximum is zero.

Theorem 5.14 ([12, p. 172 and Corollary to Theorem 1, p. 173]). *A representation-finite algebra C is biserial if and only if $\beta_R(C) \leq 2$.*

Gabriel observed that the bound by two holds for right modules if and only if it holds for left modules, as recorded in [12, p. 172]. Thus almost split sequences of right modules alone characterize the two-sided condition in Definition 4.13. To compare these sequences over S and an interval algebra, we use the interval $[0, h + 2]$: one shift contains their middle terms, and a second shift will detect which middle summands are nonprojective.

Proposition 5.15. *For the interval algebra $C = S_{h+2}$, we have $\beta_R(S) \leq \beta_R(C)$.*

Proof. Let $X \in \text{ind } S$ be nonprojective, and put $a_{YX} = \dim_k \text{Irr}_S(Y, X)$. Choose degree-one representatives of bases of all incoming irreducible spaces. Their components give a minimal right almost split map to X : Lemma 2.12 expresses every radical map as a sum of paths ending in these components, and the basis property gives minimality. Denote the graded kernel by N_X . We obtain

$$0 \longrightarrow N_X \longrightarrow \bigoplus_{Y \in \text{ind } S} M_Y \langle 1 \rangle^{a_{YX}} \longrightarrow M_X \longrightarrow 0, \quad (5.9)$$

whose underlying sequence is almost split. The target is supported in $[0, h]$, and its middle and kernel in $[1, h + 1]$, so the sequence belongs to $\text{mod } C$.

It remains almost split there. A degree-zero map that is a retraction as an ungraded map already has a degree-zero section, obtained by taking the degree-zero component of any section. Thus a non-retraction in $\text{mod } C$ remains a non-retraction over S ; taking degree zero in its almost split factorization gives a factorization in $\text{mod } C$. Right minimality also passes to this category: a degree-zero endomorphism fixing the last map in (5.9) is invertible over S , and its inverse has degree zero. The sequence is nonsplit, since its underlying sequence is nonsplit. It is therefore the almost split sequence ending at M_X in $\text{mod } C$.

If Y is nonprojective over S , apply (5.9) to Y and shift once more. Its target is $M_Y\langle 1 \rangle$, and its middle and kernel are supported in $[2, h + 2]$. This nonsplit epimorphism lies in $\text{mod } C$, proving that $M_Y\langle 1 \rangle$ is nonprojective there. Every nonprojective middle summand counted over S is consequently counted over C , with the same multiplicity. Taking the maximum proves the assertion. $\square$

To pass from biseriality to special biseriality, we use the following theorem of Skowroński and Waschbüsch.

Theorem 5.16 ([12, p. 173 and Lemma 2]; see also [10, Theorem 2]). *Every basic representation-finite biserial algebra is special biserial.*

Proposition 5.17. *If A is basic and representation-finite and $\sigma(A) = 0$, then A is special biserial.*

Proof. Choose a standard form S of A . Then $\sigma(S) = 0$ by (5.2). By Proposition 5.12, $C = S_{h+2}$ is biserial, so Theorem 5.14 gives $\beta_R(C) \leq 2$. Proposition 5.15 gives the same bound for S . The isomorphism $\Gamma_A \cong \Gamma_S$ in Theorem 5.1 preserves projective vertices and arrow multiplicities, so $\beta_R(A) = \beta_R(S) \leq 2$. Another application of Theorem 5.14 shows that A is biserial. It is therefore special biserial by Theorem 5.16. $\square$

Proposition 5.18 ([4, Theorem 3.5]). *If A is representation-finite and special biserial, then $\chi(\text{mod } A) = |A|$.*

We have proved both assertions of the magnitude conjecture.

Theorem 5.19. *Let A be a finite-dimensional representation-finite algebra over an algebraically closed field. Write $|A|$ for its number of isomorphism classes of simple modules. Then the following hold.*

- (1) $\chi(\text{mod } A) \geq |A|$.
- (2) $\chi(\text{mod } A) = |A|$ if and only if A is special biserial.

Proof. Magnitude, the number of isomorphism classes of simple modules, and special biseriality are invariant under Morita equivalence. We may therefore assume that A is basic.

- (1) Theorem 5.10 proves the inequality.
- (2) Propositions 5.17 and 5.18 prove the two implications in the equality characterization.

$\square$

APPENDIX A. DIRECT PROOFS FOR THE QUOTIENT AND ITS POSET REPRESENTATIONS

We work in Setting 3.1. This appendix proves the multiplicity, height, poset, and inequality statements used in Section 3. The proofs use the evaluation functor $F(X) = Xe$ and the τ -sequences of Proposition 3.5. They establish the required consequences of Ringel–Vossieck’s results directly in the module category.

A.1. Multiplicity one. We first bound $\dim_k \text{Hom}_A(Z, I)$ under an Ext vanishing condition. The proof uses finiteness of kernel isomorphism classes, as does the argument in [11, Section 6]; here the vanishing condition yields the following elementary lemma.

Lemma A.1. *Let A be representation-finite and I_0 an injective A -module. Suppose $Z \in \text{mod } A$ satisfies $\text{End}_A(Z) = \text{End}_A(I_0) = k$ and $\text{Ext}_A^1(U, Z) = 0$ for every submodule $U \subseteq I_0$. Then $\dim_k \text{Hom}_A(Z, I_0) \leq 1$.*

Proof. Let $f, g : Z \rightarrow I_0$ be nonzero maps with isomorphic kernels. If the kernels are nonzero, apply $\text{Hom}_A(-, Z)$ to $0 \rightarrow \ker f \rightarrow Z \rightarrow \text{Im } f \rightarrow 0$. Since $\text{Ext}_A^1(\text{Im } f, Z) = 0$, an isomorphism $\ker f \rightarrow \ker g$, followed by inclusion in Z , extends to an endomorphism of Z . The extension is a nonzero scalar, so the two kernels are equal as submodules. They are also equal if both are zero. The induced isomorphism $\text{Im } f \rightarrow \text{Im } g$ extends to a nonzero scalar endomorphism of I_0 by injectivity. Thus g is proportional to f .

Different lines in $\text{Hom}_A(Z, I_0)$ therefore have nonisomorphic kernels. There are only finitely many kernel isomorphism classes: their lengths are bounded by the length of Z , and there are finitely many indecomposables. Since k is algebraically closed, it is infinite. A space of dimension at least two over k has infinitely many lines, a contradiction. $\square$

Auslander–Reiten duality supplies the vanishing condition at the boundary of $\mathcal{D}$.

Proof of Proposition 3.9. Let Z be an indecomposable A -module. If Z is injective, or if $\tau_A^- Z$ belongs to $\text{mod } B$, then $\text{Ext}_A^1(U, Z) = 0$ for every $U \subseteq I$. In the second case, apply Auslander–Reiten duality (2.3); ordinary Hom already vanishes:

$$\text{Hom}_A(\tau_A^- Z, U) \hookrightarrow \text{Hom}_A(\tau_A^- Z, I) \cong D_k((\tau_A^- Z)e) = 0.$$

Since I is the indecomposable injective envelope of E , Lemma 3.3(2) gives $\text{End}_A(Z) = \text{End}_A(I) = k$. Lemma A.1 therefore applies and gives $d_Z = \dim_k \text{Hom}_A(Z, I) \leq 1$. Apply this argument to A^{op} and dual modules to obtain the same bound when Z is projective or $(\tau_A Z)e = 0$. As a right A^{op} -module, $D_k Z$ satisfies $(D_k Z)e \cong D_k(Ze)$. Thus duality preserves dimension at e ; it also preserves directedness and interchanges τ -projective and τ -injective objects. By Remark 3.7, every τ -injective is either injective over A or has its inverse translate in $\text{mod } B$, and dually for τ -projectives. This proves (1); the projective and injective cases give (2). For (3), the first endpoint of an almost split sequence is the translate of the last, so the two conditions reduce to (1) when the other endpoint has nonzero evaluation at e ; otherwise the required dimension is zero. $\square$

A.2. Heights. Multiplicity one gives the boundary arrows in Corollary 3.10. We now use those arrows and the paired middle terms of the τ -sequences to construct the height function. This proves directly the grading consequence of Ringel–Vossieck used in Subsection 3.4, together with its radical identities.

Proof of Proposition 3.13. Order the vertices so that the source of every arrow precedes its target; this is possible by directedness. We define the heights in that order. At P , set $h(P) = 0$. For a τ -projective $Q \neq P$, its unique incoming arrow $Y \rightarrow Q$ gives $h(Q) = h(Y) + 1$. If X is not τ -projective, then for each incoming arrow $Y \rightarrow X$, equation (3.4) gives $\text{Irr}_{\mathcal{D}}(\tau^+ X, Y) \neq 0$, so there is an arrow $\tau^+ X \rightarrow Y$. By induction, the arrow $\tau^+ X \rightarrow Y$ gives $h(Y) = h(\tau^+ X) + 1$. Set $h(X) = h(\tau^+ X) + 2$. Applying $(-)e$ to the almost split sequence over A ending at X shows that $\theta^+ X \neq 0$, so there is at least one such Y . Induction proves the asserted arrow and translation identities.

Choose maps representing bases of the irreducible spaces. By Lemma 2.12, every nonisomorphism is a linear combination of composites of these maps. Every path from X to Y has length $h(Y) - h(X)$, which proves (3.7). The nonzero maps $P \rightarrow X$ and $X \rightarrow I$ in (3.1) place all heights in the stated interval and give uniqueness at its endpoints. Every indecomposable is reached by a path from P . Since $h(P) = 0$ and height increases by one along each arrow, its height must be the length of that path. This proves uniqueness of h . $\square$

A.3. The equivalence with poset representations. Construction 3.16 describes the subspaces of Xe using maps from the τ -projectives. We prove fullness by induction and realize an arbitrary poset representation by taking a quotient of an explicit A -module.

Proof of Proposition 3.17. We verify the construction, prove fullness by induction, and then construct an A -module for an arbitrary T -space.

Step 1: Define the functor and compute it on τ -projectives. Since F is faithful and $F(Q_t), F(Q_s)$ are one-dimensional, $\dim_k \mathcal{D}(Q_t, Q_s) \leq 1$. Composites of nonzero maps between these objects evaluate to composites of isomorphisms between lines, so they are nonzero. Thus the

relation is transitive; identity maps give reflexivity, and directedness gives antisymmetry. Hence T is a poset. For $s \leq t$, normalize $w : Q_t \rightarrow Q_s$ so that $wu_t = u_s$. Then $fu_s = (fw)u_t$, proving $F_s(X) \subseteq F_t(X)$. Postcomposition by $a : X \rightarrow Y$ sends fu_t to $(af)u_t$, so G is a functor and is faithful.

Directedness gives $\mathcal{D}(Q_t, P) = 0$. Also $F_s(Q_t) = k$ exactly when $t \leq s$, giving the stated values on P, Q_t . Finally, $F(I) = k$ and $\mathcal{D}(Q_t, I) \cong D_k(Q_t e) \neq 0$, so $G(I) = K_T$. A map from $K_{\{s:t \leq s\}}$ to a T -space is determined by an arbitrary vector in its t -subspace. The map $\mathcal{D}(Q_t, X) \rightarrow F_t(X)$ is injective because $F(u_t)$ is an isomorphism and F is faithful, and it is surjective by definition. Maps from $K_\emptyset$ to $G(X)$ are arbitrary vectors of $F(X) = \mathcal{D}(P, X)$. Thus every map with source $G(P)$ or $G(Q_t)$ is induced by a map in $\mathcal{D}$, proving fullness for these sources.

Step 2: Fullness. Induct over sources in a topological order. For a non- τ -projective X , use its right τ -sequence $\tau^+ X \xrightarrow{f} \theta^+ X \xrightarrow{g} X$. Every summand of $\theta^+ X$ precedes X . Moreover, $F(g)$ is surjective: every map $P \rightarrow X$ is a non-retraction and factors through g . Given $a : G(X) \rightarrow G(Y)$, induction lifts $aG(g)$ to $b : \theta^+ X \rightarrow Y$. Faithfulness gives $bf = 0$. Since $\tau^+ X \neq 0$, this is also a left τ -sequence, so g is a weak cokernel of f . Thus $b = vg$ for some $v : X \rightarrow Y$. Surjectivity of $F(g)$ gives $G(v) = a$. This proves fullness on indecomposables, and hence on direct sums.

Step 3: Realizing every poset representation. Call an A -module K *generated at e* if $K = KeA$, equivalently if it is a quotient of a finite sum of copies of P . For each τ -projective Q of $\mathcal{D}$ and such K ,

$$\mathrm{Ext}_A^1(Q, K) = 0. \quad (\text{A.1})$$

Indeed this is immediate if Q is projective over A . Otherwise $\tau_A Q$ is annihilated by e . Since $K = KeA$, every map from K to $\tau_A Q$ is zero: the generators in Ke must map to $(\tau_A Q)e = 0$. Auslander–Reiten duality (2.3) gives $D_k \mathrm{Ext}_A^1(Q, K) = \overline{\mathrm{Hom}}_A(K, \tau_A Q) = 0$.

Take a T -space $(W; W_t)$, and form the A -module

$$V = (P \otimes_k W) \oplus \bigoplus_{t \in T} (Q_t \otimes_k W_t).$$

Here A acts on the module factor. Our chosen bases identify $Ve = W \oplus \bigoplus_t W_t$. Let $\pi : Ve \rightarrow W$ be the sum of the identity on W and the inclusions $W_t \hookrightarrow W$, and put $R = \ker \pi$. Define

$$\alpha : P \otimes_k R \rightarrow V, \quad \alpha(ea \otimes r) = ra, \quad K = \mathrm{Im} \alpha, \quad M = V/K.$$

The formula is well defined since $r \in Ve$. Since $eAe = k$, α_e is the inclusion $R \hookrightarrow Ve$, and exact evaluation gives $Ke = R$ and $Me = W$. Since K is generated at e , we have $\mathrm{Ext}_A^1(Q_s, K) = 0$ by (A.1). Applying $\mathrm{Hom}_A(Q_s, -)$ to $0 \rightarrow K \rightarrow V \rightarrow M \rightarrow 0$ shows that every map $Q_s \rightarrow M$ lifts to V . Representing morphisms of $\mathcal{D}$ by module maps therefore also gives a surjection $\mathcal{D}(Q_s, V) \rightarrow \mathcal{D}(Q_s, M)$. By Step 1, the P -summand contributes zero to $F_s(V)$, and the Q_t -summand contributes W_t exactly when $t \leq s$. Hence

$$F_s(V) = \bigoplus_{t \leq s} W_t \subseteq Ve, \quad F_s(M) = \pi(F_s(V)) = \sum_{t \leq s} W_t = W_s.$$

The last equality holds because every W_t in the sum is contained in W_s , and the term $t = s$ is present. Thus $G(M)$ is the prescribed T -space. Steps 1 and 2 prove that G is fully faithful, and Step 3 proves essential surjectivity. Thus G is an equivalence. By its definition, T has one element for each τ -projective other than P , so $|T| = |\mathrm{ind}_1^+ \mathcal{D}| - 1$. $\square$

A.4. The path-length inequality and its equality case. The arrow count in Lemma 3.14 gives $\varepsilon(\mathcal{D}) = L - |T|$. Chains of one-dimensional poset representations prove $L \geq |T|$. If equality holds, commutative squares exclude three-element antichains, and two filtrations give the required decomposition into one-dimensional representations.

Proof of Proposition 3.19. We first prove the height bound. In the equality case, we exclude three-element antichains and then decompose every poset representation into one-dimensional summands.

Step 1: The height bound. Each K_U is indecomposable, since its endomorphism ring is k . We use the equivalence G to regard it as an object of $\mathcal{D}$ and write $h(K_U)$ for the height of that

object. For upward-closed U, V , the space $\text{Hom}(K_U, K_V)$ is k when $U \subseteq V$, and zero otherwise. In the nonzero case take the map that is the identity on total spaces. Every upward-closed subset occurs in a chain from $\emptyset$ to T adding one element at a time. The inclusions in such a chain are nonzero nonisomorphisms, so each raises height by at least one by Proposition 3.13. There are $|U|$ steps before K_U and $|T| - |U|$ after it. Since $K_\emptyset = G(P)$ and $K_T = G(I)$, we obtain

$$h(K_U) \geq |U|, \quad L - h(K_U) \geq |T| - |U|.$$

Thus $L \geq |T|$ and $\varepsilon(\mathcal{D}) = L - |T| \geq 0$. If equality holds, then

$$h(K_U) = |U| \quad \text{for every upward-closed } U. \quad (\text{A.2})$$

In particular inclusions adding one element give irreducible maps, by (3.7).

Step 2: Excluding three-element antichains. We use the following observation.

Claim. If indecomposables X, Z_1, Z_2, Y , with $Z_1 \not\cong Z_2$, form a commutative square of irreducible maps as below, then $X \cong \tau^+ Y$.

$$\begin{array}{ccc} X & \xrightarrow{a} & Z_1 \\ c \downarrow & & \downarrow b \\ Z_2 & \xrightarrow{d} & Y \end{array} \quad ba = dc.$$

To prove the claim, complete the classes of b, d to bases of the incoming irreducible spaces at Y ; their sources are nonisomorphic, so these classes lie in different spaces. Representatives of these bases form a minimal right almost split map: path expansion (Lemma 2.12) gives the factorization property, and linear independence modulo the radical square gives minimality. We may therefore take b, d as components of $g : \theta^+ Y \rightarrow Y$. The nonzero tuple $(a, -c, 0, \dots)$ is killed by g , so it factors through its weak kernel $\tau^+ Y$. Hence $\tau^+ Y \neq 0$ and $\mathcal{D}(X, \tau^+ Y) \neq 0$. Both objects have height $h(Y) - 2$. By Proposition 3.13,

$$X \cong \tau^+ Y. \quad (\text{A.3})$$

Suppose t_1, t_2, t_3 are pairwise incomparable in T , and put $U = \{s : t_1 < s \text{ or } t_2 < s \text{ or } t_3 < s\}$. All sets $U \cup S$, $S \subseteq \{t_1, t_2, t_3\}$, are upward-closed. The chosen inclusion maps are identities on total spaces, so their squares commute. For $j = 2, 3$, the four objects

$$K_U, \quad K_{U \cup \{t_1\}}, \quad K_{U \cup \{t_j\}}, \quad K_{U \cup \{t_1, t_j\}}$$

form such a square, with distinct middle objects. The two squares have common source K_U and distinct targets. Equation (A.2) makes their arrows irreducible, and (A.3) gives the same translate K_U for their distinct targets. This contradicts injectivity of τ^+ .

Step 3: Decomposition into one-dimensional representations. Dilworth's theorem [6] says that a finite poset whose largest antichain has size at most two is a union of at most two chains. Apply it to T . The subspaces of a T -space W along these chains give two filtrations, after adjoining 0 and W :

$$0 = F_0 \subseteq \dots \subseteq F_r = W, \quad 0 = H_0 \subseteq \dots \subseteq H_q = W.$$

Choose C_{ij} complementing $(F_{i-1} \cap H_j) + (F_i \cap H_{j-1})$ in $F_i \cap H_j$. Induction on $i + j$, starting with the zero intersections at $i = 0$ or $j = 0$, gives

$$F_i \cap H_j = \bigoplus_{a \leq i, b \leq j} C_{ab}.$$

Indeed, put $A' = F_{i-1} \cap H_j$, $B' = F_i \cap H_{j-1}$, and $D' = F_{i-1} \cap H_{j-1}$. Then $A' \cap B' = D'$. By induction, D' is the sum over $a < i, b < j$, and

$$A' = D' \oplus \bigoplus_{a < i} C_{aj}, \quad B' = D' \oplus \bigoplus_{b < j} C_{ib}.$$

The complement of D' in A' meets B' in zero, since $A' \cap B' = D'$. Thus these three groups of summands have direct sum. Adjoining the chosen complement C_{ij} proves the next induction step. Since each indexed subspace is some F_i or H_j , bases of the C_{ij} together form a basis of W such that every indexed subspace is spanned by a subset of it. Each basis line is a one-dimensional poset

representation. Thus every T -space is a direct sum of such objects, and the equivalence proves $d_X = 1$ for every indecomposable of $\mathcal{D}$. $\square$

A.5. Three subspaces and strictness of the inequality. We give the calculations for Example 3.20. They show explicitly how a two-dimensional indecomposable contributes an extra step to the path length.

Example A.2. Let Q be the Dynkin quiver of type D_4 with arrows $1 \rightarrow 0$, $2 \rightarrow 0$, $3 \rightarrow 0$, and put $A = (kQ)^{\text{op}}$. Thus a right A -module is a diagram of three maps into its space at vertex 0. Write S_i for the simple at vertex i , and take e to be the vertex idempotent at 0, so $E = S_0$. Since 0 is a sink of Q , $P = S_0$, and the other τ -projectives of $\mathcal{D}$ are the three projectives supported on $\{0, i\}$. The poset $T = \{1, 2, 3\}$ is an antichain. We describe the indecomposables and arrows of $\mathcal{D}$ explicitly.

- (1) **Three subspaces.** For a representation V of Q , if $V_i \rightarrow V_0$ has kernel K , then V has a direct summand $S_i^{\dim_k K}$. Every indecomposable other than S_1, S_2, S_3 therefore has injective arrow maps and is represented by its three image subspaces in the space at 0.

Besides the eight one-dimensional objects K_U , $U \subseteq T$, consider

$$Z = (k^2; kv_1, kv_2, k(v_1 + v_2)),$$

where v_1, v_2 is a basis of k^2 . An endomorphism preserving kv_1 and kv_2 is diagonal; its two diagonal entries must agree if it also preserves $k(v_1 + v_2)$. Thus $\text{End}(Z) = k$, and Z is indecomposable. We show that these are all the indecomposables by decomposing an arbitrary configuration $(W; W_1, W_2, W_3)$.

A projection onto $W_1 \cap W_2 \cap W_3$ preserves all three subspaces, so it splits off copies of K_T . Once this intersection is zero, choose a projection onto $W_1 \cap W_2$ vanishing on W_3 ; it preserves all three subspaces and splits off copies of $K_{\{1,2\}}$. Repeat for the other pairs. We may now assume the subspaces have pairwise zero intersections.

Put $D = W_3 \cap (W_1 \oplus W_2)$. Choose a basis $z_j = x_j + y_j$ of D , with $x_j \in W_1$, $y_j \in W_2$, and $1 \leq j \leq m$. Both coordinate projections of D are injective, since $W_3 \cap W_i = 0$. Thus the x_j and the y_j are each linearly independent. Choose complements C_1, C_2 to their respective spans in W_1, W_2 , a complement C_3 to D in W_3 , and a complement R to $W_1 + W_2 + W_3$ in W . Since $C_3 \subseteq W_3$, its intersection with $W_1 + W_2$ lies in D ; it is zero because $C_3 \cap D = 0$. Thus

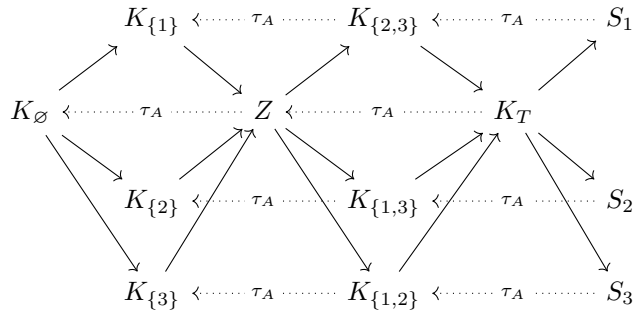
$$W = \bigoplus_{j=1}^m (kx_j \oplus ky_j) \oplus C_1 \oplus C_2 \oplus C_3 \oplus R.$$

The three subspaces decompose as

$$W_1 = \bigoplus_j kx_j \oplus C_1, \quad W_2 = \bigoplus_j ky_j \oplus C_2, \quad W_3 = \bigoplus_j k(x_j + y_j) \oplus C_3.$$

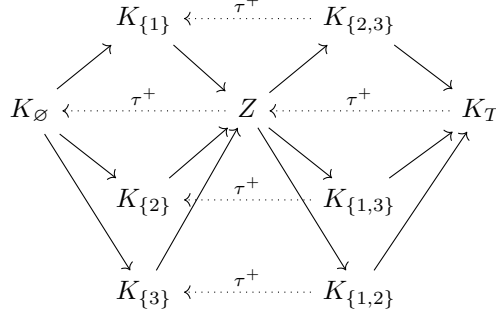
Thus each plane, with its three indicated lines, gives a copy of Z . On C_i only the i -th subspace is nonzero, and on R all three vanish. Bases of these complements therefore give copies of $K_{\{i\}}$ and $K_\emptyset$. This proves completeness.

- (2) **Irreducible maps and translation.** As A -modules, K_U has space k at 0 and at the vertices in U , with identity maps, and Z has space k^2 at 0 and maps from vertices 1, 2, 3 onto the displayed lines. Together with S_1, S_2, S_3 , these are all indecomposable A -modules. The Auslander–Reiten quiver of A is



The projectives are $K_\emptyset, K_{\{1\}}, K_{\{2\}}, K_{\{3\}}$. The almost split sequences ending at $Z, K_{\{i,j\}}, K_T$, and S_i have middle terms $\bigoplus_i K_{\{i\}}, Z, \bigoplus_{|U|=2} K_U$, and K_T , respectively. Passing to $\mathcal{D}$ sends S_1, S_2, S_3 to zero; the other objects and the arrows between them remain by Lemma 3.4.

The associated translation quiver $\Gamma_{\mathcal{D}}$ is



Index the five columns by $0, 1, 2, 3, 4$; they contain $K_\emptyset$, the $K_{\{i\}}$, Z , the $K_{T \setminus \{i\}}$, and K_T , respectively. The maps $K_{\{i\}} \rightarrow Z$ are the inclusions of the indexed lines; the map $Z \rightarrow K_{T \setminus \{i\}}$ is a quotient with kernel the i -th line. The remaining arrows are identities on total one-dimensional spaces. To compute the Hom spaces, write $L_i \subset k^2$ for the three lines of Z . Besides $\text{End}(Z) = k$, the subspace conditions give

$$\begin{aligned} \mathcal{D}(K_U, K_V) &= \begin{cases} k & U \subseteq V, \\ 0 & \text{otherwise,} \end{cases} \\ \mathcal{D}(K_U, Z) &\cong \bigcap_{i \in U} L_i, \\ \mathcal{D}(Z, K_V) &\cong \{\lambda \in (k^2)^* : \lambda(L_i) = 0 \text{ for } i \notin V\}. \end{aligned}$$

The empty intersection in the second line is k^2 . The line inclusions span $\mathcal{D}(K_\emptyset, Z)$, and the quotient maps span $\mathcal{D}(Z, K_T)$. The other maps between nonadjacent columns are scalar maps $K_U \rightarrow K_V$ with $U \subseteq V$. If $|U| = 1$ and $|V| = 2$, say $U = \{i\}$ and $V = T \setminus \{j\}$, then $i \neq j$; the inclusion of L_i followed by the quotient with kernel L_j is nonzero, so it spans $\mathcal{D}(K_U, K_V)$. All remaining inclusions across nonadjacent columns extend such a path by $K_\emptyset \rightarrow K_{\{i\}}$ or $K_{T \setminus \{j\}} \rightarrow K_T$. Maps between adjacent columns cannot factor through a third indecomposable: the same formulas permit nonzero maps between distinct objects only towards later columns. Thus precisely the twelve displayed maps form bases of the irreducible spaces. The path length from $P = K_\emptyset$ to an object in column j is j . Each solid arrow increases this length by one, and each dotted arrow decreases it by two. The nonzero τ^+ -translates are $\tau^+ Z = K_\emptyset$, $\tau^+ K_{T \setminus \{i\}} = K_{\{i\}}$, and $\tau^+ K_T = Z$. Thus $K_\emptyset, K_{\{1\}}, K_{\{2\}}, K_{\{3\}}$ are the four τ -projectives. This gives

$$|\text{ind } \mathcal{D}| = 9, \quad a(\mathcal{D}) = 12, \quad |\text{ind}_1^+ \mathcal{D}| = 4, \quad \varepsilon(\mathcal{D}) = 18 - 12 - 4 - 1 = 1.$$

- (3) **A path of length greater than $|T|$.** The map $K_{\{1\}} \rightarrow K_{\{1,2\}}$, an identity on total spaces, factors as

$$K_{\{1\}} \xrightarrow{1 \mapsto v_1} Z \xrightarrow{(a,b) \mapsto a-b} K_{\{1,2\}}.$$

The second map kills the third indexed line. Both maps preserve the indexed subspaces. The height differences in the chain $K_\emptyset \rightarrow K_{\{1\}} \rightarrow K_{\{1,2\}} \rightarrow K_T$ are $1, 2, 1$. Thus $L = 4$ whereas $|T| = 3$, giving $\varepsilon(\mathcal{D}) = 1$, while the indecomposable Z has total dimension two.

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